Overall Project title:

Developing data science solutions to mitigate the health impacts of climate change in Africa: the HE²AT Center

The Heat and Health African Transdisciplinary Center (HE²AT Center)

Study Protocol

Innovative machine learning and multi-source data analysis towards development of an urban heat-health Early Warning System for African cities

Protocol number: HEAT002

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Harnessing Data Science for Health Discovery and Innovation in Africa

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Acronyms

Acronym	Definition
ACT	Antenatal Corticosteroids Trial
AES	Advanced Encryption Standard
API	*Application Programming Interface*
ART	*Anti Retroviral Treatment*
BP	Blood Pressure
CDC	Centers for Disease Control
Co-PI	Co Principal Investigators
CSAG	Climate Systems Analysis Group
DASH	Data and Specimen Hub
DMAC	Data Management and Analysis Core
DMP	Data Management Plan
DoH	Department of Health
DRS	Data Reference Syntax
DSA	Data Sharing Agreements
ECMWF	European Center for Medium range Weather Forecasts
EDCTP	European and Developing Countries Clinical Trials Partnership
EWS	Early Warning System
EWSs	Early Warning Systems
FIC	Fogarty International Center
FTP	File Transfer Protocol
GANTT	Generalized Activity Normalization Time Table
GCM	Global Climate Model
GCRO	Gauteng City Region Observatory
GSOD	Global Summary Of the Day
HDSS	Health and Demographic Surveillance System
HIV	Human Immune deficiency Virus
HPC	High Performance Computing
LST	Land Surface Temperature
MCV	Mean Cell Volume
MODIS	Moderate Resolution Imaging Spectroradiometer
MRC	Medical Research Council
NDVI	Normalized Difference Vegetation Index

NIST	National Institute of Standards and Technology
NIH	National Institute of Health
OLI	Operational Land Imager
PI	Principal Investigator
POPIA	Protection of Personal Information Act
RHI	Reproductive Health and HIV Institute
RoB	risk of bias
RP1	Research Project 1
RP2	Research Project 2
RPs	Research Projects
SC	Steering Committee
SCL	Scene Classification
TEC	Training and Engagement Core
TIRS	Thermal Infrared Sensor
TOA	Top Of Atmosphere
TWC	The Weather Company
UCT	University of Cape Town
UPGC	University Peleforo Gon Coulibaly
USAID	United States Agency for International Development
UW	University of Washington
WHC	Wits Health Consortium
WHO	World Health Organization
WMO	World Meterorological Organisation
WWARN	Worldwide Antimalarial Resistance Network

SECTION 1: OVERVIEW AND DESCRIPTION OF INVESTIGATORS AND INSTITUTIONS

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University of Washington	United States of America	Center for Health and Global Environment (CHanGE), Department of Environmental and Occupational Health Sciences, School of Public Health, University of Washington, Box 354695, Seattle, Washington, United States of America
University Peleforo Gon Coulibaly	Cote d'Ivoire	UPGC BP 1328 Korhogo, Côte d'Ivoire / CSRS, 01 BP 1303 Abidjan 01, Adiopodoumé- Km 17-Route de Dabou, Côte d'Ivoire
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Synopsis

Rationale: The study constitutes one of two Research Projects (RPs) within the NIH-funded HE²AT Center. It specifically addresses the complexity of urban spaces with regard to heat-health impacts and the appropriate responses for some particular vulnerable groups.

Objectives: The overarching goal is to advance understanding of complex spatially and demographically stratified heat-health interactions in large African cities and to apply this information to develop locally relevant and risk-stratified Early Warning Systems (EWS). The aims are three: (1) Map intra-urban heat vulnerability and exposure across urban areas in large African cities (Aim 1); (2) Develop a spatially and demographically stratified heat-health outcome forecast model in order to predict the probability of adverse health outcomes at different temperature thresholds (Aim 2); and (3) Develop an Early Warning System reflective of geospatial and individualized risk patterns (Aim 3)

Study design: The RP2 focuses on the conditions in two large cities in two regions of Africa (Johannesburg, SouthAfrica, Southern Africa, and Abidjan, Côte d'Ivoire, West Africa). It adopts a transdisciplinary approach in which multidisciplinary experts will collaborate with communities, local government actors and policy makers to address heat-related complex and interconnected research gaps. Existing data from longitudinal studies (trials and cohorts) in the two cities that were performed among HIV-infected adults, HIV-uninfected adults and adults in COVID-19 prevention or treatment studies will be reanalysed, together with weather, other environmental and socio-economic and other data. Analysis will deploy a range of machine learning methods to construct an index of intra-urban socio-economic and environmental vulnerability factors. As the solutions developed by the RP2 will address two major concerns for global policymakers (how to warn people about a heatwave in urban settings in low- and middle-income countries, and then to track its impacts), local, national and international policymakers will be engaged at all stages.

Dissemination: Findings will be disseminated at various levels, using several channels, including workshops, policy and research fora, scientific conferences and journal publications, and towards different target groups (including policy makers, communities, specific vulnerable groups).

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Keywords: urban, heat, health, early warning systems, intra-urban vulnerability, socio-economics and environment, exposure mapping.

SECTION 2: RESEARCH PROTOCOL

1) Background and rationale

1.1 High ambient temperatures and heat waves

High ambient temperatures above long-term averages during summer months and discrete heat extremes (e.g., heat waves) are associated with excess mortality and considerable morbidity [1-4] . WHO predicts that by 2030 there will be almost 92,000 deaths per year from heat waves, with sub-Saharan Africa amongst the worst affected regions [5]. Anthropogenic climate change has already resulted in a more than 1°C rise in temperature globally since the Industrial Revolution [6]. This increase is not evenly distributed across the planet, however, or even within local areas [7].

Regional differences and the effect of urban development and land use change means that many parts of Africa are experiencing higher than average temperature increases, and more frequent, intensive, and longer-lasting heat waves [8]. Work done by a team at the University of Witwatersrand estimated that temperatures in Africa will rise at about 1.5 times the global rate of increase, escalating by 4-6°C above current temperatures in the sub-tropics by 2100 [8]. Heat waves have been defined in a myriad of ways, based on static or relative thresholds (e.g. maximum or mean temperature, and taking into account humidity and even solar radiance), number of days of excessive heat, and timing during the summer season.

1.2 Specific urban heat vulnerability

The 'Urban Heat Island' is a phenomenon where concrete, non-reflective surfaces, and low greenery and wind results in temperatures considerably higher than in surrounding areas, and concomitant high levels of morbidity and mortality during heat waves [7]. This is concerning as Africa is the most rapidly urbanizing continent in the world and, by 2050, 59% of its population is expected to live in cities (up from 40% in 2018) [9].

In many African cities, nearly half of the population lives in informal settlements or "slums" that are often located in areas unsuitable or undesirable for other uses [10]. People living in these areas have high exposure to heat, as, for instance, informal settlements are often in low-lying hotter parts of cities, are typically densely settled and crowded, lack vegetation cover, have limited or no shade, and poor natural ventilation [11]. Moreover, typical housing materials in informal housing – iron metal sheeting – accentuate heat exposure, with temperatures inside these dwellings commonly 3-4°C warmer than outdoors [12, 13].

In fact, much of the urban environment in African cities offers little heat insulation, with low-cost government-built housing, school classrooms and prefabricated clinics, for example, also often exceeding outdoor temperatures by 3-4°C, or more [13-16] . People inside these structures commonly report heat-exposure symptoms such as fatigue, difficulty breathing, and headaches [14, 16-18]. Heat exposure in occupational settings such as manual labour in factories, construction sites or other outdoor activities can also reach dangerous levels [19, 20].

1.3 Vulnerable groups and spaces in urban contexts

Urban poor are most vulnerable to heat exposure for a range of factors related to heightened sensitivity (such as high rates of HIV, malnutrition, and non-communicable diseases) and to lowered adaptive capacity (such as lack of access to cool water, air-conditioned spaces, health services, and

occupational protections). Taking all these factors together means that, for example, an elderly woman living in a tin informal dwelling who has respiratory compromise from previous TB will have very few or no options for protecting herself against heat stress. We need to understand the impact that a heat wave has on her health outcomes, and what more can be done to protect her and other vulnerable groups in our warming world. This is critical information as most heat-related morbidity and mortality is preventable with improved preparedness and avoidance of exposure.

Impacts of heat exposure are clearly contingent not just on temperature, but also on geographical, socioenvironmental, and demographic factors [20]. Therefore, robust quantifications of the magnitude and pattern of heat vulnerability, exposure, and impacts, need to include an assessment of the extent to which meteorological parameters (temperature, rainfall, wind direction, and humidity, for example), urban forms, population density, and other factors contribute to risk. Importantly, the unique characteristics of the environmental exposures, built environment, demographics, and spectrum of communicable and non-communicable diseases in Africa mean that findings elsewhere may have little applicability to African cities.

To date, however, none of the assessments of heat-health impacts in African cities has taken adequate account of the complexity of urban spaces, and most have relied on relatively small datasets with poor spatial and temporal coverage. Considering the spatial heterogeneity of living conditions in cities in Africa, mapping intra-urban heat vulnerability and exposure across urban areas is of utmost importance.

1.4 Health outcomes: heat related morbidity and mortality

Worldwide, the majority of studies of heat hazards in cities quantify environmental vulnerability and develop temperature hazard maps without linking these to actual health outcomes (see for example: [21-23]). Equally, studies that focus primarily on health outcome data such as excess mortality during heat waves seldom integrate data on vulnerabilities from the built environment or urban space [24, 25]. Many studies use summary health outcome data for a district, or hospital record data [25, 26], and assume that the temperatures that patients experienced in the district and their living conditions are uniform. Coarse location information constrains the ability to map health outcomes, and limits the extent to which outcomes may be linked with environmental stressors and population activities such as work and residential location.

Studies of heat impacts in cities in high-income countries have found that risk for heat-related morbidity and mortality are highest among the elderly, those with chronic cardiovascular or respiratory conditions, and people who are homeless or in lower socio-economic groups [19]. Some studies in Africa have documented higher heat health risks among the elderly [23, 24, 26-29], pregnant women [27-29], and children under five years [24, 30-34]. Most studies, regardless of setting, have not considered vulnerability as determined by geospatial factors representing the built environment form.

It is also important to note that the majority of heat impact studies in Africa and worldwide relied on analyses of temperature associations with mortality because these data are often most easily accessible. However, from a public health perspective, it is more important to understand heat-related *morbidities* as these account for a considerably greater burden of disease than heat-related mortality. Additionally, from an evidence viewpoint, there are major gaps in evidence on heat-related morbidities than mortality.

1.5 People with particular health problems

Little evidence is available on whether high temperatures pose particular risks for some of the key vulnerable population groups in sub-Saharan Africa, such as those living with HIV. This is a critical question: there are an estimated 26 million people living with HIV on the subcontinent, up to 30% of adults are infected in some countries on the sub-continent, and some 440,000 deaths are due to AIDS annually [35, 36]. Of note, some recent studies have found that rates of HIV are highest in areas most vulnerable to heat impacts, such as informal settlements [35] This is a critical question: there are an estimated 26 million people living with HIV on the subcontinent, up to 30% of adults are infected in some countries, and some 440,000 deaths are due to AIDS annually [35, 36]. Of note, some recent studies have found that rates of HIV are highest in areas most vulnerable to heat impacts, such as informal settlements [35] . In a PubMed (MEDLINE) search on 7/11/2020, we located three studies that had documented the impacts of heat exposure (measured through actual temperatures) on disease patterns in HIV-infected people [37], and 13 that had identified increased levels of adverse conditions during the summer or the 'warm' seasons [38-50]. Increased rates of adverse health conditions during the warm seasons strongly suggests that more nuanced analyses of temperature exposures will demonstrate sizable temperature-outcome relationships and identify actionable information.

Reducing risks to health from current and projected high temperatures depends not only on physiological acclimatization, but also on planned adaptation by the health and other sectors. Early Warning Systems, one of the key adaptative interventions applied across the globe, is discussed hereafter.

1.6 Early warning systems (EWS)

Early Warning Systems (EWSs) play a central role in the health sector response to heat waves*[51-57]*. These EWS form the central element in the health system's response to heat waves [58, 59].

In these systems, once a predefined temperature threshold is exceeded, warnings trigger a set of actions intended to lower exposure to harmful temperatures and to assist people to take protective actions. Common interventions in high-income countries include opening of public cooling shelters, targeted messaging, wellness checks in the community at people's homes or in institutions such as homes for the elderly, distribution of cold water, and surveillance systems [58, 60]. EWSs have been shown to reduce morbidity and mortality during heat waves [52, 61], including in low- and middle-income countries [62].

EWSs in Ahmedabad, India for example, led to a dramatic decline in heat wave deaths in 2015 compared to a similar heat wave in 2010[62]. Over longer time scales, changes to the built environment such as green roofs, shade structures, reflective surfacing that can reflect infrared and UV wavelength and reduce heat transfer, as well as thermal emittance. Reflective surfacing includes painting surfaces white can reduce heat exposures in urban areas [63, 64]Reflective surfacing includes painting surfaces white can reduce heat exposures in urban areas [63, 64]. However, these structural interventions are unlikely to be feasible at scale in most African cities for some time. Well-functioning, adequately sensitive EWSs accompanied by preventive measures such as dedicated cool spaces for high-risk groups, community health worker outreach, water supplies, and closing of schools and

workplaces will be the mainstay of heat wave responses in the short- to medium-term in African cities. Over longer time scales, changes to the built environment such as green roofs, shade structures, reflective surfacing, and painting surfaces white can reduce heat exposures in urban areas [63, 64]. However, these structural interventions are unlikely to be feasible at scale in most African cities for some time. Well-functioning, adequately sensitive EWSs accompanied by preventive measures such as dedicated cool spaces for high-risk groups, community health worker outreach, water supplies, and closing of schools and workplaces will be the mainstay of heat wave responses in the short- to medium-term in African cities [61].

Most EWSs have cut-off thresholds directed to the entire population and do not take the level of vulnerability of different groups into consideration [65]. There is growing evidence that most heat-related illnesses and deaths occur on dates without heat warnings [66], suggesting that current thresholds could be improved to take particularly vulnerable populations into account.

1.8 Research gaps and needs of a multi-disciplinary approach

Within Africa there is a pressing need to identify:

- the aspects of socio-economics and demographics, such as housing types and density, commuting distances, and working conditions, that contribute to heat vulnerability and exposure
- the types of heat exposure that are most dangerous for people living in different conditions in African cities (e.g., maximum in night-time or daily temperatures, heat waves or heat spells of long duration combined air quality & heat/cold snaps)
- geospatially detailed heat-hazard conditions across cities related to altitude, vegetation, water bodies, and risk groups under different large-scale weather conditions
- people's and in particular most vulnerable groups' perception and understanding of heat as a health hazard and how these risks are framed

This research study will contribute to providing answers to these questions that would constitute a major leap towards developing interventions to reduce vulnerability and build resilience against extreme heat in African cities. It will adopt a transdisciplinary approach, drawing together knowledge in the fields of biomedical and climate science, as well as geospatial analysis and computer science.

2) Aims and Objectives

The overarching objective of this Research Study is to advance understanding of complex spatially and demographically stratified heat-health interactions in large African cities and to apply this information to develop locally relevant and risk-stratified Early Warning Systems(EWSs).

More specifically, this overarching objective will be pursued through three aims:

1. Map intra-urban heat vulnerability and exposure across urban areas in large African cities (Aim 1)
    a. Quantify intra-urban socio-economic and environmental vulnerability (Aim 1a)
    b. Develop high resolution urban temperature hazard maps (Aim 1b)
2. Develop a spatially and demographically stratified heat-health outcome forecast model to predict the probability of adverse health outcomes at different temperature thresholds (Aim 2)
3. Develop an Early Warning System reflective of geospatial and individualized risk patterns (Aim 3)
    a. Develop an EWS based on a digital App (Aim 3a)
    b. Develop a "tiered" EWS in collaboration with the Ministries of Health, National Meteorological and Hydrological Services, and other relevant stakeholders (Aim 3b)

The methods for Aim 3 are not presented in this protocol and will be elaborated in a subsequent protocol.

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3) Summary of overall study design

This study adopts a transdisciplinary approach in which experts in the fields of computer science, healthcare, climate science, statistics, physics, physiology, environmental epidemiology, and public health will collaborate with communities, local governance actors and policy makers to address heat-related complex and interconnected research challenges, in African urban contexts.

The study will pursue the three aims in an integrated and a consistent way with a logical flow of activities from understanding the local urban contexts to the development of forecasting models intended for the development of an effective and appropriate urban early warning system for heat.

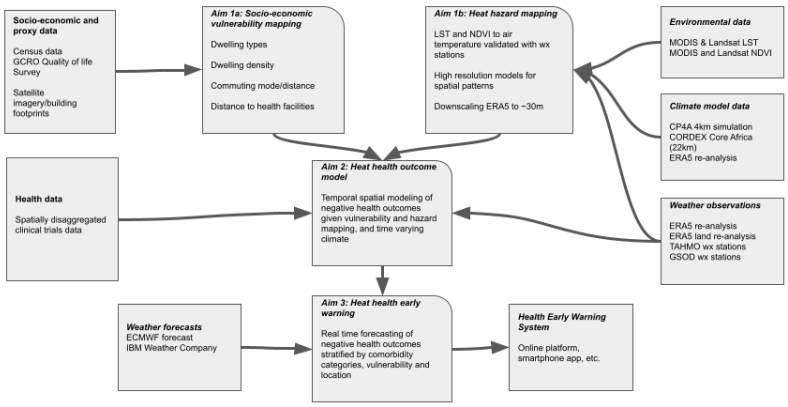


Figure 1 Flow of activities from Aim 1 to Aim 3, starting from an analysis of past observations, to the implementation of forecasting and downscaling methods to generate an Urban Heat-Health Early Warning System.

The study will make use of existing, prospectively collected, high-quality health data from longitudinal datasets (see in Annex 1 Table 4 the description of health outcomes, environmental and other datasets). Additionally, the study will make use of a multitude of existing weather data, both retrospective or historical, as well as weather and seasonal prospective or forecast data (see table4 in Annex 1).

The study will depend on a DMAC (Data Management and Analysis Core) hosted at the University of Cape Town. DMAC will implement a remote access analysis platform based on Python Jupyter Lab which is a collaborative data science coding platform enabling access by study partners to the High-Performance Computing (HPC) and large storage provided by UCT. Climate System Analysis Group (CSAG) at UCT hosts a HPC cluster and large storage system that already holds many of the relevant climate datasets with a number of new geospatial datasets being made available in support of the HE²AT study.

The study will also draw on the resources of IBM-PAIRS (Physical Analytics Integrated Repository and Services), which is an extensive data store of aligned, pre-processed geo-spatial data such as satellite and aerial images, Light Detection and Ranging (LiDAR) data, land use surveys, numerical weather prediction model outputs, weather reanalysis and measurement, and other environmental data, Internet of Things(IoT) sensor data, anonymized cell phone tracking based human mobility and social-economic data such as population, age-distribution, capacities of health infrastructure, and road networks.

To date, PAIRS has curated more than 6 petabytes(PB) of data at a daily ingestion rate of about 20 terabytes(TB). The technology is based on a large-scale Hadoop/HBase system for distributed data storage and processing. IBM-PAIRS leverages efficient data indexing methods, which result in spatially and temporally linked data layers, both for data from 2Dimensional grids (e.g. satellite images) and from point locations (e.g. measurements from distributed sensor networks).

Access to PAIRS will either be directly through the various PAIRS interfaces which will be made available by providing authentication to identified individuals, or through integration into the DMAC data analysis platform (Jupyter Hub) which will also involve authentication for identified individuals. Full details of data access and safety are in Annex 6.

4) Methodology

4.1. Study sites

The study covers two large cities in two regions of Africa (Johannesburg, SouthAfrica, Southern Africa, and Abidjan, Côte d'Ivoire, West Africa).

Johannesburg is the largest city in South Africa, classified as a megacity, the 26th-largest in the world, with 14,167,000 inhabitants in 2021[67]. Johannesburg is in the eastern plateau area of South Africa known as the Highveld, at an elevation of 1,753 metres. It has a subtropical highland climate, with the summer months (October to April) characterised by hot days followed by afternoon thundershowers and cool evenings, and the winter months (May to September) by dry, sunny days followed by cold nights. Temperatures in Johannesburg are usuall mild due to the city's high elevation, with an average maximum daytime temperature in January of 25.6 °C (78.1°F), dropping to an average maximum of around 16 °C (61°F) in June. The city is a Köppen Climate Classification subtype Cwb, with January the warmest month, with an average temperature of 20.6°C (69.0°F).

Abidjan is the largest city in Ivory Coast, with a population of 4.7 million (estimation in 2014), which is 20 percent of the overall population of the country, and this also makes it the sixth most populous city proper in Africa. Abidjan lies on the south-east coast of the country, on the Gulf of Guinea. The city experiences a tropical wet and dry climate. In Abidjan, the wet season is warm and mostly cloudy, the dry season is hot and partly cloudy, and it is oppressive year-round. The hot season lasts for 5.7 months, from November 22 to May 13, with an average daily high temperature above 30.5°C (87°F). The hottest month of the year in Abidjan is April, with an average high of 31°C (88°F) and low of 25.5°C (78°F). The cool season lasts for 2.5 months, from July 3 to September 19, with an average daily high temperature below 28°C (82°F). The coldest month of the year in Abidjan is August, with an average low of 23°C (73°F) and high of 27°C (81°F). Abidjan is a Köppen Climate Classification subtype Am: Tropical Monsoon Climate, with February the warmest month with an average temperature of 28.3°C (83.0°F)[68].

Given the marked differences between these two cities, the study will be able to undertake cross-region comparisons and assess whether variations between sites can be explained by differences in vulnerability, heat hazards or in underlying disease patterns between the two areas. Having one coastal, tropical city, and one inland, high elevation, sub-tropical city will provide a valuable basis to test the transferability of models and methods in different geographic settings.

4.2. Major steps and integration

The study will be conducted following four major steps presented in the Figure 2 below. These steps correspond to the study aims.

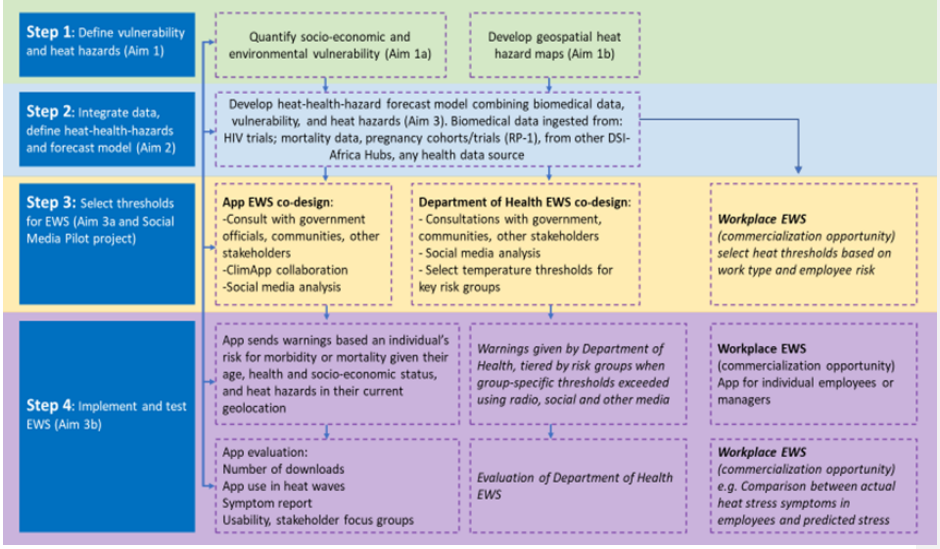


Figure 2 Steps for achieving the aims and objectives of the study

The research activities over the two cities will be conducted in a collaborative and an integrated way, including the teams from the two cities working together, sharing experiences and expertise in team building and capacity-building, and attending workshops organised in one setting or the another.

Details of the work from data acquisition to data processing and data analyses are described below.

4.3. Data description and acquisition

The different data types of data and datasets are presented in Annex 1

, and described below under each aim and sub-aim, as relevant.

4.3.1. For Aim 1a: Quantify intra-urban socio-economic and environmental vulnerability

For Aim 1a, none of the data that will be used are identifiable human health data. All social or demographic and health data that will be used are available publicly and only at an aggregated level. The data privacy and related ethical concerns that apply to analysis of human health data therefore do not apply to data used in this aim.

Two categories of data will be used.

The first category is socio-economic survey data. These data will provide information on household economic status or living standards, demographic and environmental characteristics.

- City observatory surveys data (e.g., in South Africa, Gauteng City Region Observatory (GCRO) survey data including the Quality-of-Life survey (GCRO QoL survey 2020/21), and in Abidjan the Bureau National d'Etudes Techniques et de Development Reports including the Urban Slums Reports and similar data sets)
- Population Censuses and other community surveys (e.g., in South Africa, Stats SA census and community surveys such as Census 2011 and the Community Survey 2016, and similar datasets in Ivory Coast)

The second category consists of proxy spatial measures of socio-economic conditions:

- Satellite image-based estimation of land use, land cover types, building and roof types (GCRO/Terra Image 2012).
- Access to services including electricity
- Proximity to health care centres and services (e.g., Johannesburg's public hospital and clinics directory, and similar data sets in Abidjan, Ivory Coast)
- Access to green spaces (e.g., City Parks directory, and similar data sets in Abidjan, Ivory Coast))
- Other (e.g., Google Street View analysis, distributions of human population, such as SEDAC-gridded human population data)

The satellite imagery and other geospatial data acquired in Aim 1a is used to characterize urban vulnerabilities to heat and for mapping heat- or temperature-related hazards. For example, the satellite image views of the city will be classified to capture different typologies (e.g., slum, industrial, green areas, and suburban residences), where this information will then be used to account for differences in the diffusion of heat (in Aim 1b). The Stats SA small area layer from the 2011 Census is an example of socio-economic geospatial dataset that may be used for identifying urban residential areas and their characteristics such as dwelling types and the type of access to services electricity and water, which may be useful when quantifying and mapping heat hazards.

The data for achieving Aim 1a is freely available, or already curated within the existing IBM or University of Cape Town databases.

4.3.2. For Aim 1b: Develop high resolution urban temperature hazard maps

For Aim 1b, only weather and climate data will be used for this aim (no human data or measurements). The data privacy and related ethical concerns that apply to analysis of human health data therefore do not apply to data used in this aim.

For urban temperature, several datasets will be used for downscaling which will subsequently also be used for developing an urban-level Early Warning System. The weather datasets we will use can be summarised into two categories.

The first category, **retrospective or historical data**, will serve as proxy ground truth data entailing observations or proxy observations for temperature across the case study urban areas (e.g., Johannesburg and Abidjan). Included in this are model-based re-analysis data, which use physics and numerical weather simulations in combination with observations from a global network of sources (e.g., weather stations, satellites, and simulation models). Examples of these datasets include:

- MODIS and LANDSAT Land Surface Temperature (LST)
- UK Met Office/IMPALA CP4-A (4km resolution 10-year climate simulation over Africa)
- TAHMO weather stations (https://tahmo.org/)
- WMO Global Summary Of the Day (GSOD) weather stations
- ECMWF ERA-5 Re-analysis of regional circulation fields (zonal and meridional winds, temperature, humidity near surface (10m and 2m above the surface) as well as on relevant pressure levels)
- ECMWF ERA5-Land which provides the output of a higher resolution (10km) resolution land surface model forced by ERA5 and includes variables such as LST

This category of data will serve as a reference of what happened in the past in terms of weather and climate. Reanalysis datasets, such as ECMWF's ERA5 and ERA5 Land have a global spatial extent with hourly temporal resolution and coverage starting from 1950 to the present, which is extremely valuable for machine-learning studies which require sufficient temporal coverage to perform model training and evaluation. ERA5 reanalysis data consists of several climate variables, such as 2m temperature, sea-surface temperature, relative humidity, soil temperature, and total precipitation, all of which can be used as additional sources of predictability when post-processing and de-biasing physics-based weather/climate forecasts with machine learning.

The second category of data consists of outputs from dynamical physics-based **forecast or prospective data** emanating from global or regional ocean-atmosphere coupled climate models:

- ECMWF sub-seasonal to seasonal forecasts
- NOAA Climate Forecast System (v2) sub-seasonal to seasonal forecasts
- IBM The Weather Company seasonal forecasts
- NOAA NCEP GFS short-range forecasts

Dynamical short and medium-range forecasts, like the ones listed above, will assist in developing an urban heat-health early warning system based on sound physical properties. ECMWF issues 1–46-day ensemble forecasts at daily intervals for horizons of 1-14 days ahead, and twice a week for horizons from 15-46 days ahead. NOAA CFSv2 issues a 90 day ahead horizon forecast every day. IBM's The Weather Company issues a 1-7 month ahead daily forecast every month. All of these forecast products are available operationally from the IBM PAIRS platform at various spatial resolutions and are accessible via an API. These forecast datasets can be pre-processed on demand in PAIRS on using map-reduce functions in order to be harmonized with climate datasets hosted by CSAG. The spatial resolution of most of these gridded forecast products varies between 10km to 50km.

The data for achieving Aim 1b is already curated within the existing IBM or University of Cape Town databases.

4.3.3. For Aim 2: Develop a spatially and demographically stratified heat-health outcome forecast model

Aim 2 incudes reanalysis of existing data on human health. Data privacy and related ethical concerns that apply to analysis of human health data are considered below and further unpacked in Section 6 below.

The datasets used in study activities for this aim are drawn from existing longitudinal datasets collected in the studies conducted in Johannesburg, South Africa or in Abidjan, Côte d'Ivoire. This encompasses data from trials and cohorts among HIV-infected adults, HIV-uninfected adults, and adults participating in longitudinal studies related to COVID-19 prevention or treatment. We will include data from all participants 18 years and older at the time of enrolment in the primary studies, and not use sampling to select particular sub-sets of individuals.

Data acquisition will occur through:

a) approaching the Principal Investigators of potentially eligible datasets to request data sharing and their participation in the research project
b) approaching data repositories that store the relevant data
c) completion of legal agreements on data transfer between the study team, and the data provider or data repository, as relevant

The following databases will be included:

a) datasets from studies done within research facilities within the City of Johannesburg or Abidjan geographical area, regardless of whether participants live outside the city boundaries
b) databases that contain the population groups of interest, without restrictions on the size of database
c) studies that enrolled participants since the year 2000
d) the investigators have given permission for use of the data

Though the geolocation data on the area where individual participants reside is the optimal level of geolocation that we require, we will not exclude databases where participant geolocation data is not available. In those instances, we will use the geolocation of the research site where participants attended study visits as a proxy for their geolocation.

We will include both interventional and non-interventional studies. Longitudinal studies include cohorts where participants are enrolled and then followed over time.

We will select datasets to include based on convenience sampling, where we balance considerations around:

a) the number of datasets we can analyse within the available financial and human resources available for the study
b) the size of the study and duration of follow-up in a study
c) the number and types of health variables collected
d) the quality of the study and the data
e) the availability of participant geolocation data
f) data sharing stipulations within the study (e.g., whether data sharing was explicitly included in informed consent, or if data sharing was mandated by study funders)
g) interest of the study investigators in sharing data

We will include data from three categories of trials/cohorts, each done in different study populations. Drawing data from three different categories of studies aims to ensure that the size of the datasets are sufficiently large for the training of machine learning models, but another valuable reason for the inclusion of data from different population groups is that this provides an opportunity to examine sub-group level associations, and thus compare heat-health outcomes across groups.

The first category of studies is those done among people with HIV infection. Most of these studies were done to assess the effectiveness or efficacy of different antiretroviral treatment drugs. These data will allow us to investigate specific associations between heat and other environmental exposures and health outcomes in HIV-infected people. The associations between heat and health outcomes in HIV-infected people are likely to differ in size and nature from heat-health linkages in other population groups.

The second category of trials or cohorts is those done among populations at risk for acquiring HIV infection (HIV-negative adults). These are mostly trials that were done to assess the effectiveness or efficacy of interventions to prevent HIV acquisition. Interventions include vaginal microbicides and pre-exposure prophylaxis with antiretroviral drugs.

The third category of studies that will provide data for re-analysis will be sought from COVID-19 prevention or treatment trials conducted in primary study sites of Johannesburg[69-71]. These may include trials of COVID-19 vaccines. Studies done on COVID-19 followed large cohorts of people both with and without HIV, providing a rich data set to map against the weather and other data collated in Aims 1a and 1b. Further, these longitudinal COVID-19 studies have been done in recent years when there have been clearer provisions for data sharing.

Variable categories which will be drawn from these studies are presented in Annex 3 (Table 5). These include variables related to social, economic and demographic characteristics, geospatial location of participants, self-reported health status, medical examination findings, laboratory outcome data and outcomes of other types of investigations are presented in Annex 3 (Table 5).

The use of the personal information of study participants places this study under the restrictions and permissions described by the Protection Of Personal Information Act of South Africa (POPIA 2013). The requirements of the act with regards to processing of personal information, accountability, security and quality, informed consent, and subject notification, are detailed in the associated Data Management Plan[72].

4.4. Data pre- and post-processing, formatting

The following sections describe the data processing steps specific to this research study. The over-arching data management plan is described in detailed in the associated DMP document.

4.4.1 For Aim 1a: Quantify intra-urban socio-economic and environmental vulnerability

- The data will be aggregated/interpolated onto the same spatial units (such as the Stats SA Census 2011 small area units, or GCRO QoL survey spatial units), and saved as ESRI Shapefile(s) or equivalent (GeoJSON/PostgreSQL-postGIS)
- Image-based data aggregation to spatial units will include mean values and counts, or proportions, as well as variance to represent the heterogeneity within each spatial unit (e.g., mix of *township formal* and *township informal*)

No identifiable human data are used in Aim 1a activities (only data aggregated to small levels, and not human data at an individual level).

4.4.2 For Aim 1b: Develop high resolution urban temperature hazard maps

- Data will be processed to align with the CSAG Data Reference Syntax (DRS) which is based on international DRS such as CORDEX and CMIP. This provides a uniform structure for directories and filenames.
- Climate and Forecasting (CF) conventions will be applied to ensure file level meta-data conformity. Data will be stored primarily in NetCDF (NetCDF4-HDF5) format through subsets software and will enable access to particular data in other formats (e.g., CSV).

There is no human data involved in Aim 1b activities.

4.4.3 For Aim 2: Develop a spatially and demographically stratified heat-health outcome forecast model

- The health datasets will be harmonised into a single coherent dataset through codebook remapping on the variables listed in Annex 3.
- Codebook remapping entails:
    - Step 1 taking the variables across the different datasets
    - Step 2 feeding the data into the study database
    - Step 3 defining the variables on a continuum which allows for all data to be

  For example, pre-term delivery might be translated from "number of days" to an ordinal categorical variable such as the trimester of preterm birth. Another example occurs when measuring levels of education. In some cases, the number of years attended may be the measure as opposed to situations where total subjects passed would need to be put on the same continuum to be useful for analysis.
- Throughout the pre-processing process, steps will be taken to deidentify data to ensure patient privacy is adhered to. These steps are described in full in section and the associated DMP .
- We will undertake post-processing based on machine-learning and a forecast skill evaluation as described in the next sub-sections.

4.4.3.2 Machine-Learning Based Post-Processing

- It has been shown by various studies that machine learning has the ability to improve the forecast skill of dynamical models at a global and regional scale[73-75]. In order to obtain the best possible forecast of temperature and other relevant climate variables, machine learning will be used to post-process and blend various dynamical forecasts. This approach will aim to de-bias dynamical forecasts to better account for rare or extreme events that will be of particular interest for the urban early warning system. Approaches that can be implemented include algorithms like Xgboost or Ngboost, convolutional, and recurrent neural networks.

4.4.3.3 Forecast Skill Evaluation

Evaluating the skill of a forecast product in the vicinity of the regions of interest is necessary to establish what the expected performance will be in an operational setting, especially as a function of horizon (lead time). The following baseline forecasts are normally used to evaluate the skill of a new forecast product:

- Persistence - which is obtained by simply taking the most recent available observation at a given location
- Climatology - which is obtained by taking a long-term average (~30 years) of a given variable, at a specified location/pixel and for a given time-of-year. This represents the climate that can be expected at a certain point on the globe for the considered time-of-year.

More advanced climatologies that take trends into account (as a result of climate change) also serve as a good forecast baseline. The goal for a new forecast product is to improve on these baselines. For Machine Learning-based post-processed forecasts, the goal will be to improve on the skill of the baselines as well as the base dynamical model(s) on which post-processing methods are being applied, as well as climatology. Emphasis will be given to evaluating forecasts for their ability to predict extreme events and events categorized in e.g., the 70th, 80th, 90th percentile ranges. Evaluation metrics that will be used will include Mean Absolute Error (MAE), Anomaly Correlation Coefficient (ACC), and Ranked Probability Skill Score (RPSS), among others.

Once an ML-based post-processed GCM forecast product has been developed for lead times of between 1-30 days with sufficient skill, downscaling approaches will be applied to obtain urban-level estimates of temperature and other relevant variables, conditioned on the coarse GCM forecast. This forecast will be an integral component for the Urban Heat-Health EWS. Figure 2 shows an overview of the components related to downscaling of temperature maps and how the resulting high-resolution maps will interface with the EWS.

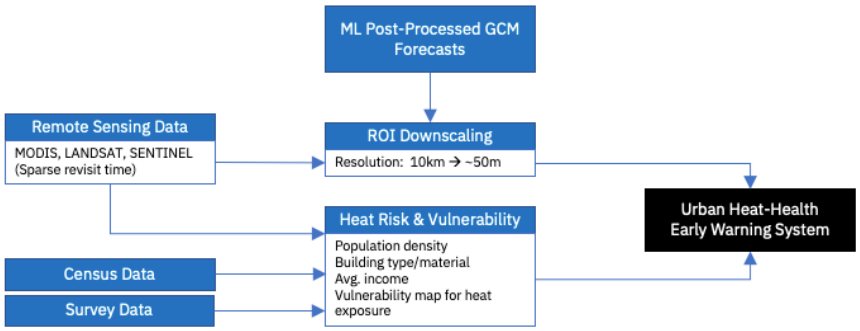


Figure 3 Overview of the data and workflow for downscaling and how it will be used in the Urban Heat-Health Early Warning System

4.5. Data analyses and outputs

4.5.2. For Aim 1a: Quantify intra-urban socio-economic and environmental vulnerability

- Dimensionality reduction methods will be implemented, for example the use of Principal Component Analysis (PCA) to identify dominant correlation structures across the variables. This will produce candidate predictors or features that best explain intra-urban socio-economic variability as well as a way to combine the components into a single indicator that confers combined socio-economic and environmental vulnerability that may be associated with a wide range of climate risks, but in our contexts, this would be primarily heat. Hierarchical or multi-level clustering methods can also be applied to identify regions similar in

terms of vulnerability. A critical consideration in terms of methods, will be accounting for spatial variability, hence the appropriate approach would be to implement techniques such as spatial principal component analysis and geodemographic clustering methods. Aggregation into a single output map will be implemented using spatial multi-criteria analytic method.

4.5.3. For Aim 1b: Develop high resolution urban temperature hazard maps

The analyses will mainly include downscaling.

- MODIS and LANDSAT LST data will be processed to fill in gaps resulting from cloud cover on a day-to-day basis. Different gap filling approaches will be explored including auto-regression and multi-variate regression imputation using coarse scale temperature estimates from ERA5 and/or ERA5-Land.
- Land surface temperatures are a derived product from the Landsat multi-spectral band information. Landsat 8 Top Of Atmosphere (TOA) 16-bit images which are available at a 16-day temporal interval. The Operational Land Imager (OLI) sensor provides multispectral bands 1-7 and 9 with 30 m spatial resolution and Thermal Infrared Sensor (TIRS) which collects data at a 100 m spatial resolution (bands 10 and 11). One of the main algorithms that will be implemented makes use of the thermal infra-red band and band 5 & 7 [76]. Typically, the resulting resolution using this algorithm is 100 m for LST, but with the requirement to produce higher spatial resolution temperature grids for correlation with health outcomes, more complex approached will be pursued aimed at 30 m resolution LST. One of these will build upon the work of Mushore et al, 2022, who developed a multivariate regression method that used several land cover indices (e.g., Normalized Vegetation Index (NDVI)) and topographical variables (elevation and slope) to downscale LST to 30 m resolution. Based on this, a statistical downscaling model will be developed, trialling a number of machine learning approaches, to predict sub-daily (night time minimum and day time maximum) spatially coherent local scale (~30 m resolution) near surface air temperature patterns across the city. Recent methods such as the random forest regression kriging or quantile random forest regression kriging methods [77] will be investigated. Generative models, specifical variational auto-encoders are candidates that will also be considered for this task.
- Physics based models will be used to derive near surface air temperature from LST data based on the physical disaggregation of latent and sensible heat fluxes. Machine learning approaches will be compared to the physical approach.

4.5.4. For Aim 2: Develop a spatially and demographically stratified heat-health outcome forecast model

- Data analysis done as part of Aim 1a will be extended under Aim 2, in the development of the heat-health outcome model, where geolocated socio-economic variables will be used as predictors and machine learning models (e.g., the quantile random forest model which enables identification of important variables and a measure of uncertainty) will identify the strongest predictor variables from the suite of socio-economic variables
- Machine learning models will be developed and tuned (various supervised approaches will be trialled, such as regression, decision trees, random forest, K-Means, support vector machines) that capture the predictive relationship (associative) between negative health outcomes and temperature.
- Examples of health impacts that are of interest include:

    - Impact on other health outcomes: High blood pressure, respiratory stress, cardiac conditions. HIV related blood tests such CD4 counts, basophils PRT, eosinophils PRT, MCV (Mean Cell Volume), lymphocytes PRT
- Predictors that will be considered for this model will be required to be geographically coincident (to the limits of the health geolocation data) and will include downscaled near surface air temperature estimates, estimates of socio-economic conditions (e.g., dwelling type, commuting conditions, etc.) and more preferably the indicator of socio-economic vulnerability to heat.
- An important component to consider in the models that will be developed will the discovery of sub-groups or risk strata. As an initial step, different manual aggregations will be explored based on inputs from biomedical or epidemiological experts.
- Participants in the databases will be stratified into different sub-groups, such as age category and socio-economic status. As a follow-up step, the applicability of automatic sub-group discovery methods will be investigated, and these include the use of open-source tools such as Pysubgroup and auto-stratification tools that have been developed by IBM.
- Models will be trained to maximize predictive skill relative to different groups in order to identify models that provide the most skill for outcomes of the greatest value.
- The results of machine learning analyses need to be considered in terms of existing evidence of plausible biological or social pathways, for example. Engagement with health practitioners and researchers will be done to provide feedback on the relative value of different models, and selection of optimum algorithms.

5) Special data management considerations

5.1. Data types

Multiple categories of data will be used across the project divided into four categories.

5.1.1. Health related data

Health related data for this study will focus on clinical trials datasets generated by clinical trials undertaken in the case study cities (Johannesburg and Abidjan). Trials datasets will be identified based on the scale of the trial as well as the availability of geospatial variables (e.g. clinic locations or other geospatial information) in order to allow spatial mapping of health outcomes and the intersection with socio-economic spatial mapping and climate variable spatial mapping.

Health related data is generally considered to be personal data as it pertains to individuals medical records, diseases history and health events. In some cases health related data is also considered *personally identifiable data* where individuals can be identified either directly through names, ID numbers, etc. or indirectly deduced from address, GPS locations, or cross referencing with other identifiers. Many countries now have specific laws dictating conditions for sharing and processing of such data and associated protective measures to avoid unauthorized access and potential harm to individuals involved in the studies. In South Africa the relevant legislation is the Protection of Personal Information Act (POPIA 2013). This necessitates processing in compliance with the specific

requirements of the POPIA act. These requirements and compliance with them is outlined in detail in the section below: POPIA compliance

Due to above mentioned privacy, ethical and legal requirements, health related data acquisition almost always involves an associated DSA between the data provider and the HEAT Center (see data management workflow description below). In particular, where health data is traveling across international borders, compliance with the relevant national laws, permissions, and associated requirements for the sharing and transfer of health data will be followed.

Once data is in South Africa, local ethical clearance and, where data constitutes personal information, POPIA compliance apply.

5.1.2. Climate/weather data

These data include both observational-based datasets (weather station observations, satellite proxy observations, and processed/gridded observations). Gridded climate data produced from atmospheric re-analysis and climate simulations will form historical gridded climate observations and forecasts.

Climate-related data will in almost all cases involve accessing open data repositories such as Copernicus Climate Data Store (CDS), or Earth System Grid Federation data systems. Climate-related data will be stored on IBM Research Africa and CSAG/UCT systems and during the transition period on WHC-managed cloud services with auto-scaling capabilities for handling variable seasonal data volumes and improved accessibility for authorised researchers, with CSAG/UCT responsible for managing and updating the relevant data indexes. These datasets also follow open data policies, typically requiring citation when used for non-commercial research.

Deleted: will be stored on IBM PAIRS data storage and/or CSAG/UCT data storage systems. however, CSAG/UCT will manage and update the primary data index for climate-related data available to the consortium.

Remote sensing data excluding climate data

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Remote sensing data are collected via satellite-based sensors that capture a wide range of environmental variables. These datasets enable large-scale and continuous monitoring of factors such as land surface temperature, soil moisture, vegetation condition, land use/cover, building density, and cloud cover. Remote sensing is particularly valuable for analysing areas where ground-based observations are sparse or unavailable, and it complements other types of climate and environmental data.

Commonly used satellite datasets include those from **MODIS, Landsat, and Sentinel missions**, along with products available from **open repositories such as the Copernicus Climate Data Store (CDS)** and **Sentinel data systems**. In most cases, these datasets are made available under open data policies with no restrictions on non-commercial research use, though citation of the original source is required.

All remote sensing–related data will be stored on **IBM Research Africa** and/or **CSAG/UCT systems**, with **CSAG/UCT responsible for managing and updating the relevant data indexes**. During the transition period, data may also be stored on **WHC-managed cloud services**, which offer auto-scaling capabilities to handle variable seasonal data volumes and improve accessibility for authorised researchers.

Remote sensing data will not be used to identify individuals in any way and do not constitute sensitive or personally identifiable data.

Areal/Geospatial socio-economic data

Socio-economic data represent measures of household and community conditions, including household economic status, access to services such as water and sanitation, and dwelling type. Typical sources include national census data and focused household or demographic surveys. These data are **not sourced through remote sensors**.

Socio-economic data will be obtained from both open and restricted-access repositories (e.g., South African census data and GCRO Quality of Life Surveys). Primary copies will be indexed and stored on **CSAG/UCT data storage systems**. In addition, versions may be uploaded to **IBM Research Africa systems** to enable processing and analysis. The IBM Research Africa team will acquire access to some of the geo-referenced Quality of Life survey data from GCRO, process these data on their internal computing systems, and share the resulting outputs and indices with the rest of the HE²AT Center Consortium through the **HE²AT Center's Data Access Committee (DAC)**.

South African census data are already available through the **UCT DataFirst repository**, while GCRO Quality of Life Survey data are available through the **GCRO open data platform** as well as via direct queries to GCRO.

Both the South African census data and the GCRO Quality of Life survey data are aggregated to small areas and therefore do not constitute personally identifiable or sensitive data.

5.2. Data indexing and processing (harmonization, homogenization, normalization)

Data indexing and processing (harmonization, homogenization, normalization) will be facilitated through both the CSAG/UCT platform and WHC-managed cloud infrastructure with enhanced processing capabilities and automated backup systems. The cloud infrastructure provides GPU acceleration for complex machine learning algorithms and real-time data processing essential for Early Warning System development.

CSAG currently implements a **Data Reference Syntax (DRS)** which is a structured mapping from a controlled vocabulary of meta-data elements to a directory and file naming syntax. This is standard practice within climate data management and we will continue to implement this approach for the climate and remote sensing datasets. To maintain transparency, all versions of mappings and data transformations are controlled through CSAG's GitLab system. This version control ensures that any changes to the data are documented, and previous versions can be restored if necessary

Deleted: The focus is on data derived from satellite sensors, mainly optical imagery (e.g. satellite images of urban centers) as well as indicators of physical measures such as land surface temperature, soil moisture estimates, vegetation condition, land use and cover, etc.) for the purposes of estimating environmental quantities, and land use/building density etc. Remote sensing data will not be used to identify individuals in any way and does not constitute sensitive or personally identifiable data. Climate data is often collected through remote sensing. However, for the purposes of this study, we will classify climate data collected through remote sensing as in the first category of data (Climate/weather data), and not in this category. ¶
Remote sensing related data will in most cases involve accessing open data repositories such as Copernicus Climate Data Store (CDS), Sentinel data systems. Remote sensing related data will be stored on IBM PAIRS data storage and/or CSAG/UCT data storage systems. CSAG/UCT will manage and update the primary data index for remote sensing related data.¶
Most remote datasets used are available through open data policies with no restrictions on non-commercial research use. We will cite the original source.¶

Deleted: These data represent measures of socio-economic and related conditions such as household economic status, access to services such as water and sanitation, dwelling type, etc. Typical sources include national census data, and more focused household & demographic survey data. These data are not sourced through remote sensors. ¶
Socio-economic data will be sourced from both open data repositories as well as restricted access repositories (e.g. South African census data, and GCRO Quality of Life Surveys). Primary copies will be indexed and stored on CSAG/UCT data storage but versions may already exist or will be uploaded on IBM PAIRS to enable analysis through IBM PAIRS.¶
South African census data is already available through the UCT DataFirst data repository and GCRO QoS data is available through the GCRO open data platform as well as through direct queries with GCRO.¶
South African census data is aggregated up to small areas and so does not constitute personally identifiable data. GCRO Quality of Life survey data is likewise aggregated to small areas and does not constitute personally identifiable sensitive data.

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Deleted: Data currently available on the CSAG/UCT data storage system as well as data available on IBM PAIRS will be indexed using an appropriate pragmatic meta-data standard and this index will be made available by CSAG/UCT data platform (Gitlab).

The current DRS is documented on the CSAG Gitlab Wiki which is accessible only to team members with requisite credentials.

Integration with DSI-Africa Open Data Science Platform will ensure that meta-data propagates to the ODSP system.

Environmental data (climate data, remote sensing data, socio-economic mapping data) will be homogenized to align with existing meta-data and storage standards (CF conventions and OGC standards) as documented on the Gitlab Wiki.

Health data harmonization and de-identification will require meta-data/codebook mapping from source codes through to common codes (to be developed). Codebook mapping will involve both RP and DMAC participation but will be technically implemented by DMAC. The codebook remapping will be developed and documented separately to this DMP.

5.3. Data management, documentation and curation

Data management will be facilitated through both the CSAG/UCT platform and WHC-managed cloud infrastructure with enhanced security and automated backup systems. Database population utilises cloud-native database services, providing automated backup and recovery systems with enhanced data integrity and disaster recovery capabilities essential for continuous Early Warning System operations.

Deleted: Data will be managed by a team of data specialists within the study and ***primary data management*** will take place on the CSAG/UCT data platform.

IBM PAIRS will continue to implement their own internal data management; however, this will include sharing of meta-data indexing with CSAG/UCT for data related to NIH HE²AT Center and DS-I Africa.

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Individual partner institutions will have their own data and computing platforms ranging from central institutional platforms through to personal computing devices. Data management within these partner institutions does not fall under this data management plan and we refer to this as ***secondary data management***. However, partners will be encouraged to regularly consider what locally managed data should be integrated into the study wide data management and made available to the broader DSI-Africa programme and beyond.

Primary data management will involve:

- Homogenization to agreed data archive standards (see above for each class of data)
- Version control of datasets that are regularly updated to ensure prior version remain accessible
- Harmonizing of health-related data
- Meta-data indexing within the CSAG/UCT DMP as well as DS-I Africa ODSP
- Documentation of data on CSAG/UCT GitLab wiki
- Sharing of data management code through CSAG/UCT GitLab code repositories and WHC-managed cloud infrastructure

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5.4. Data Security and Confidentiality of Potentially Sensitive Information

The use of health datasets requires careful consideration of data security and confidentiality. This is strongly guided by the relevant legislative context for each datasets including the specific country

legislation around the use of personal/sensitive data and the cross-border transfer of such datasets. The development and negotiation of these data sharing agreements.

In order to support the development of the Data Sharing Agreements and ensure compliance with legislation around access to and protection of personal/sensitive information the following security measures will be implemented:

5.4.1. De-identification

POPIA Section 10 prescribes the principle of Minimality which means that only information relevant to the purpose of processing should be processed. Where personal information is acquired that is required to fulfill the research purposes described by the relevant research project protocols, de-identification will be implemented according to the following protocol which is guided by US Department of Human and Health Services (HSS) guidelines[78].

1. Street addresses may be aggregated into geographical regions such that many records map to the same region and it is no longer possible to derive individual residential locations. For example, in RP2, where the highest spatial granularity is needed to map urban heat health outcomes, census small areas or wards with spatial scales of the order of 2 to 5 km will be used to aggregate records. Aggregation will also consider the number of records that map to the same geographical area.
2. Latitude/longitude coordinates may be "jittered" which involves adding random values to each coordinate such that the exact location is lost[79] assesses different approaches to location masking and jittering), while retaining sufficient geographical information to support the analysis. This will likely be the case for RP1 where jittering will be used to shift latitude/longitude locations in the order of 10s of kilometers, adequate to prevent locating individual residential locations.

Geo-location masking/jittering and aggregation will be considered adequate through expert determination involving experts from UCT, IBM and NIH.

Data encryption

Data is transferred using Transport Layer Security (TLS) protocols are employed during transmission to maintain encryption and prevent interception. We use WeTransfer modality, which is encrypted. Once transferred to UCT and during the transition period to WHC-managed cloud infrastructure, any data identified as containing Personal Identifiers or specified by the DTA is encrypted for storage using AES-256 encryption standard for both UCT servers and cloud storage. Encryption key access is restricted to authorised personnel, ensuring data security and preventing re-identification attempts. This complies with privacy and data protection legislation, including the Protection of Personal Information Act (POPIA). Metadata, however, is stored separately to facilitate indexing and software development whilst maintaining security.

Storage isolation

Primary encrypted data will be stored in an isolated virtual storage server that is only accessible to private data team.

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Deleted: Primary data (before de-identification) will be encrypted using 256bit AES (Advanced Encryption Standard) a standard established by the US NIST (National Institute of Standards and Technology) with encryption keys only available to the minimum number of people required to implement the de-identification processing once data is transferred to the UCT HPC system. ¶
The use of cryptographic modules validated to NIST FIPS 140-2 is required by the United States Government for encryption of all data that has a classification of Sensitive but Unclassified (SBU) or above. We will use these standards when transferring and storing data. ¶
For datasets that include personal identifiers (ie. personally identifiable data), datasets will be encrypted by the original data provider and transferred to UCT through a secure data transfer that uses TLS encryption.¶

5.4.2. Network firewall and Virtual Private Network

The CSAG compute infrastructure benefits from UCT's comprehensive security policies and during the transition period from WHC-managed cloud security architecture with 24/7 Security Operations Centre (SOC) monitoring and advanced threat detection systems for improved security posture and compliance. Key measures include:

• Firewall Protection: UCT's Cisco firewall and cloud-native firewall services safeguard against external threats, ensuring only authorised access is permitted.

• VPN Access: A Cisco VPN service encrypts all traffic, enabling secure remote access to the UCT intranet and maintaining confidentiality.

• Access Control: Access to CSAG servers and services is carefully managed, with strict limits on authorised users.

5.4.3. Local authentication and authorization

Data is transferred using Transport Layer Security (TLS) protocols are employed during transmission to maintain encryption and prevent intereception. We use WeTransfer modality, which is encrypted. Once transferred to UCT and during the transition period to WHC-managed cloud infrastructure, any data identified as containing Personal Identifiers or specified by the DTA is encrypted for storage using AES-256 encryption standard for both UCT servers and cloud storage. Encryption key access is restricted to authorised personnel, ensuring data security and preventing re-identification attempts. This complies with privacy and data protection legislation, including the Protection of Personal Information Act (POPIA). Metadata, however, is stored separately to facilitate indexing and software development whilst maintaining security.

Beyond UCT's broader security measures, the CSAG/UCT platform employs additional authentication and authorisation protocols alongside cloud-specific access controls and automated compliance reporting for streamlined governance oversight during the transition period. User identities are verified through a Linux filesystem and Lightweight Directory Access Protocol (LDAP), with access to restricted datasets managed through UCT's authentication protocols and internal CSAG Data Management Plan mechanisms.

All activities comply with UCT's information security policies, ensuring adherence to institutional standards.

By implementing these comprehensive encryption modalities, network security and authentication measures, the HE²AT Center ensures the protection and confidentiality of sensitive data throughout its lifecycle, maintaining compliance with ethical and legal standards.

From the start of Year 5, enhanced cloud security measures include:

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UCT broader security policy:
http://www.icts.uct.ac.za/sites/default/files/image_tool/images/286/UCT_Information_Security_Policy_PC03_2020.pdf¶
UCT ICTS implements a Cisco firewall and Cisco Virtual Private Network service to ensure that intranet access is limited to authorized users.¶
UCT firewall policy
http://www.icts.uct.ac.za/uct-perimeter-firewall-policy¶
¶
These network level security measures ensure that access to the UCT intranet is controlled to authorized UCT users. Furthermore, CSAG server access is controlled by CSAG authentication and authorization as described below.¶

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• AES-256 Encryption: All data at rest and in transit

• 24/7 SOC Monitoring: Continuous security monitoring and threat detection

• Role-based Access Controls: Multi-factor authentication and granular permissions

• Automated Backup Systems: Enhanced disaster recovery capabilities

• Compliance Monitoring: Automated reporting for regulatory compliance

5.5. Procedure for making data available to qualified individuals

The Steering Committee (SC) will develop a Data Access Request Form which people requesting data need to complete prior to consideration of their request. The form will include a proposal outline of the intended research, and the procedures to maintain data confidentiality and security.

The SC will review and approve or reject all requests from the research community, which includes scientists or medical professionals working at academic, non-profit or government institutions, or commercial companies. We will ensure that all requests conform with NIH policies and procedures including compliance with informed consent procedures if relevant, and any limitations stipulated by the institutions/investigators who contributed data to either of the RPs. The study will comply with the principles of the Data Protection Act of the country of the participating site. Some countries may have restrictions on data sharing outside of the host country, which we will abide by. Researchers who request to share the resources of the HE²AT Center will need to agree to not seek to identify individuals within the dataset, not distribute the data to any other entity, keep the data secure, and acknowledge the HE²AT Center and DS-I Africa as appropriate in publications and presentations (the exact acknowledgement text to be agreed among the DS-I Africa Program). Further, researchers who share the Center's resources will be strongly encouraged to collaborate with and train African investigators as part of the work they conduct with the resources shared.

There may be exceptions to the resource sharing plans outlined above. Firstly, there may be considerations around intellectual property protections for the research products that the consortium aims to commercialize. Decisions about resource sharing in these circumstances will follow the NIH policies in this regard, including those on resources sharing, disseminating unique research resources, and program income. The NIH will be provided with a copy of documents or sample of these products developed under the grant award (e.g. the Digital App). A large portion of the data we will use in the Project activities will be drawn from the IBM-PAIRS platform, which contains several datasets that require a license for access. We will thus not be able to share those datasets without permission from the licensee.

Clearly, the rights and privacy of individuals who participate in research must be protected at all times. Thus, data intended for broader use should be free of identifiers that would permit linkages to individual research participants and variables that could lead to deductive disclosure of the identity of individual subjects. All data shared beyond the HEAT Center will be de-identified following the procedures described above. Requests for original data containing personally identifiable or sensitive information will be referred back to the original study.

The HE²AT Center team will especially endeavour to make the unique research resources that we develop readily available to other researchers working on climate and health. There is a pressing need within the field to expedite the translation of data into knowledge and, in particular, into interventions that protect people against extreme heat and other manifestations of climate change. We thus feel obligated to expedite our responses to such data requests. Fellow researchers wishing to access the data will thus not have to wait for our research findings to be accepted for publication or for the 'final research data" prior to data sharing, provided the research questions they are exploring do not directly overlap with the specific question we are addressing.

The study involves 'data sharing' on a large scale, where we are reliant on the willingness and ability of data owners to share. In that spirit, as the holders of the database it will be beholden on us to share the data with other groups, provided the original data providers agree to sharing and the procedures of HE²AT Center data sharing processes are followed. We will develop formal collaboration agreements around data sharing with the data owners who contribute data to the database, which set out the terms and conditions for data reuse. This is important, as for example, some of these investigators may not wish their data to be shared with third parties, for example. While a willingness to share data forms the basis of this project and most often reflects a desire of investigators to collaborate, this may not necessarily translate into a willingness to then share data beyond the Center. Sharing of the datasets gathered into the database will thus require signed agreement from the original data owners and may require an application to the local IRB where the original study was done. People who make use of the database will have to undertake to adhere to the authorship guidelines stipulated in the collaboration agreement signed between the HE²AT Center and the research groups who contributed their data.

Of note, much of the data and resources generated by the Hub will be useful for a range of other disciplines. The research resources generated in this study such as vulnerability-heat-health data visualization will also potentially have a wide range of applications, among people working on urban geography or planning, for example. Additionally, the datasets have tremendous potential to answer a wide range of questions, outside of environmental health. We will facilitate access to these resources by interested parties. We also undertake to share data and resources generated by the HE²AT Center with other Hubs or components in the DS-I Africa program, wherever possible. In particular, the DS-I Africa Open Data Science Platform will be used as a mechanism for making relevant datasets from the HE²AT Center available across the program, and beyond.

Data and data resources (primarily software/code) will be shared through standard protocols such as File Transfer Protocol (FTP) servers and the UCT instance of Gitlab (software version control system). UCT also hosts an open data portal which will be used to make more final datasets such the urban heat vulnerability maps available following FAIR principles. Data that is made publicly or otherwise available through data use and sharing agreements will be accompanied by metadata such as data dictionaries and data descriptors such as principal investigator, funding sources, data collector, project description, sample and sampling procedures, temporal and geographic coverage of the data collection, variables, technical information on files (file formats, linking, etc.), interviewer guides and coding instruments.

Analytic data sets will be provided as de-identified data files that can be read by common statistical package software, such as SAS or STATA. De-identified data sets will have names and other personal health identifiers removed (see section 5). Data and other resources will be transferred to others under the terms of a DSA to ensure that the data will be used for the proposed purpose and that no attempts will be made to identify participants. We will maintain records of all researchers who have been given access to the research resources. Results of the HE²AT Center activities will be shared

with the research community and the public through conference presentations, publication in peer-reviewed journals and media interactions, as described in the Training and Engagement Core(TEC).

According to the Research Data Management policy, 'publicly funded research data are a public good, produced in the public interest, which should be made openly available with as few restrictions as possible in a timely and responsible manner'. Data is therefore open by default, closed by exception (e.g. privately-funded research, or research with commercialisation possibilities).

5.6. Principles of data sharing and open access

5.6.1. Restrictions to data sharing

According to Section 4.6 of the UCT Research Data Management policy,

> "[n]ecessary constraints on the availability of data include the protection of personal data; the protection of intellectual property; the protection of commercial interests of study partners; and security concerns."
>
> Strategies to limit restrictions may include data being anonymised or aggregated; gaining participant consent for data sharing; and gaining copyright permissions.

5.6.2. Discoverability

The HEAT Center and DMAC will implement FAIR principles to ensure that:

- Data will be **findable** through publicly accessible and searchable meta-data indexes The DS-I Africa ODSP and UCT's ZivaHub repository both offer platforms for metadata searching.
- De-Identified data will be **accessible** via a data access request to the DAC, which, if approved, will require a Data Transfer Agreement.
  - **Inter-operability** will be enabled through the strict adherence to established data and meta-data standards (see above)
  - **Reuse** will be supported through rigorous documentation of the data including limitations and guidance for reuse.

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Deleted: either openly through a public facing component of the DMP data repository, or through a data access request to DMAC (where a Data Sharing Agreement is required)

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6) Ethical and Privacy considerations

Ethical considerations and legal requirements

Summary of general ethical and legal requirements

The project raises several ethical and legal considerations. This includes 1) the use of secondary data for research purposes that may differ from the original purpose of the study, 2) sharing data across countries in Africa, and 3) the presence of potential identifying information.

The project will endeavour to ensure the protection of human subjects and their data through procedures that adhere to the following legal and ethical guidelines:

- Declaration of Helsinki
- ICH Good Clinical Practice
- Ethics in Health Research (Department of Health South Africa) (109)
- Protection of Personal Information Act (POPIA)
- Any additional African country-specific protection of data legislation

Table 1 below summarises some of the ethical and legal considerations for the project, and how these will be addressed. Additional details describing these procedures are described thereafter and further detail included in the DMP in Annex 6.

Table 1: Summary of ethical, legal and data management considerations that underpin study activities

Ethical considerations	Legal requirements	Data management activities
• The purposes of the research are clearly defined in the study protocol and the reasons that the study needs to involve personal data are provided • Processing of historical data for research purposes in the study serves the public interest • Specific concerns of vulnerable populations such as pregnant women, neonates, and children have been considered • We assess ethics concerns in the studies that contribute data, including information on the ethics committee that provided approval for the original study, securing a copy of the informed consent of the study, if possible, and clarifying any concerns about the study with the data provider, where feasible • Participant informed consent has been addressed, separated into two scenarios: ○ Participant has signed a ''broad consent'' for use of their data in future research projects, even though the precise nature of the future research is unclear, this will	• POPIA requirements are met • The data provider has determined that the country's legal requirements for personal data security are adhered to • Data sharing agreements are in place for each dataset from data provider to UCT • Data sharing agreements specify the use of the data through UCT's database by other partners like IBM, CeSSHAR and Wits RHI • Any time limitations on use of data are adhered to	• A data management plan clearly defines how the data is transferred, stored and secured • Clearly defined and limited personnel will have access to data through access-controlled platforms, with specific individuals authorized for different levels of data access (e.g. only specific named individuals have access to encryption codes, and a log of all those who have access to the data for analysis) • Only data that is necessary for the conduct of the study will be collected and data will be anonymised, when required • Publications will not include any potentially identifiable information from participants, and only aggregated data will be published or presented publicly

allow data sharing without further ethical approvals ○ Participants that have signed ''narrow consent'', where sharing of data is not permitted, and/or "tiered consent", where each individual participant chooses to permit data sharing or not, will be carefully considered: ▪ The possibility for reconsenting will be deliberated if no consent for data sharing has been provided by the participant. In cases where this is feasible (e.g. ongoing studies), reconsenting is done ▪ If reconsenting is not feasible, impossible, or would involve a disproportionate effort, an informed consent waiver will be requested from the **Wits Human Research Ethics Committee** • Ethics committee has approved the conduct of the study and are provided with information on each of the studies that contribute data • Risks to individuals' privacy and potential for identification are limited as far as possible		

Assessment of the informed consent process in primary study

The informed consent process for each dataset that agrees to contribute data to the study will be assessed for ethical considerations. Firstly, we will review informed consent procedures that were undertaken by the participant for the primary study. If a participant has signed a ''broad consent'' for use of their data in future research projects, even though the precise nature of the future research is unclear, this will allow data sharing without further ethical approvals.

Participants that have signed ''narrow consent'', where sharing of data is not permitted, and/or "tiered consent", where each individual participant chooses to permit data sharing or not, will be carefully considered. The possibility for reconsenting will be deliberated if no consent for data sharing (narrow and tiered with no consent) has been provided by the participant. If reconsenting is not feasible, impossible, or would involve a disproportionate effort, an informed consent waiver will be requested from Wits Human Research Ethics Committee.

Waivers of consent for use of data for purposes other than that stipulated in the informed consent form

Waivers of informed consent are relatively common in public health settings. For example, in instances where data are used on individuals or groups to prevent the spread of communicable diseases, as occurred in the COVID-19 pandemic. Oftentimes health information is collected without consent to determine the incidence of various health conditions. This is analogous to our study where we wish to calculate the incidence of adverse health outcomes during hot periods, compared to cold periods. The justification for these breaches of the normal right of individuals to informed consent is that this right can be subordinated to the needs of the community if the two are in conflict. Naturally, the extent to which this subordination applies to research needs to be determined for each research proposal.

The informed consent requirements for participation in research are stricter than those for both individual and public health interventions because the research participants are exposed to the risks of research but often do not receive any benefits even when the research is successful. In contrast, public health interventions, such as programmes to reduce the impacts of extreme heat on health, often do benefit the recipients of the intervention. In the absence of a waiver, the research would be severely impeded, or not possible in fact.

We acknowledge that for research using identifiable human data, researchers must normally seek consent for the collection, analysis, storage and/or reuse. The Declaration of Helsinki, paragraph 25 notes that it is preferable that potential research participants be informed in advance about any possible future uses of their data so that it will not be necessary to re-contact them when another such use is proposed. When the information in databases are anonymized and aggregated, it is often impossible to identify the source persons from whom to ask informed consent for new uses. This is especially true for databases that were established in the past when informed consent for future uses was not sought. Research ethics committees may decide to waive the informed consent requirement by considering the Declaration of Helsinki's exception to the requirement as stated in paragraph 25: "There may be situations where consent would be impossible or impractical to obtain for such research or would pose a threat to the validity of the research. In such situations the research may be done only after consideration and approval by a research ethics committee." The CIOMS International Ethical Guidelines for Epidemiological Studies (commentary on Guideline #4) also provides support on this exception.

POPIA compliance and protection of personal information

The use of health datasets requires careful consideration of data security and confidentiality. This is strongly guided by the relevant legislation context for each datasets including the specific country legislation around the use of personal/sensitive data and the cross-border transfer of such datasets.

The Protection of Personal Information Act of South Africa (2013) institutes limitations on the processing of personal information, but also provides the legal basis for the use of personal information for the purpose of scientific research. POPIA functions alongside other legislation and regulatory structures governing research in South Africa, such as the Constitution of the Republic of South Africa, the National Health Act No 61 of 2003 and regulations as prescribed by the Minister of Health, such as the South African Department of Health guidelines on Ethics in Health Research Principles, Processes and Structures 2015 ("DOH Guidelines"). The law which takes precedent will be that which provides the most comprehensive protections to the rights of individuals in South Africa.

Section 6 of the Act indicates that the Act does not apply in cases where personal information has been de-identified to the extent that it cannot be re-identified again. Where this is the case, the information can be used without restriction or conditions of the Act. It is envisaged that a number of health databases acquired for the project will have been de-identified to the extent that re-identification is virtually impossible. However, where this is not the case, the following sections of POPIA provide a basis for processing health data.

While Section 15(1) requires that personal information processing must be compatible with the purpose of collection, Section 15(3)(e) allows for processing for the purposes of historical, statistical and research purposes regardless of the original purpose of collection. As the study will be using health datasets collected prior to the project, we will never be processing personal information compatible with the original purpose of collection, unless specific consent for further processing was agreed to by the data subjects. Section 15(3)(e) therefore provides the basis for processing information where this specific consent was not obtained.

Furthermore, while Section 18(1) lays out requirements for informing the data subject that their personal information is being processed for a particular purpose, Section 18(4)(f) provides for an exemption from these requirements where the information will be used for historical, statistical or research purposes.

Furthermore, Section 14(2) allows for the retention of personal information for research purposes as long as safeguards are in place to prevent the use of the personal information for other purposes

Section 19 requires that security measures are implemented to prevent the unlawful access to or processing of personal information. We will implement extensive data security measures. Section 20 describes requirements around individuals operating on or processing personal information. We have identified the individuals who will be responsible for this processing and will continually update this list. Access to personal information is restricted through passwords and other security measures so that only individuals authorized by the responsible party have access.

Section 21 requires a written contract between the responsible party and the operators implementing processing. This contract will be signed by all individuals processing personal information and is presented in the data management plan and includes agreement that the operator will inform the responsible party where there is any basis for believing that personal information has been accessed by an unauthorized person.

As noted in Section 27 of the POPIA Act, in instances where the study investigators are requesting access to data for a purpose which was not originally agreed upon by the participant in the informed consent, we will only use the data for research and statistical purposes.

We believe that the inclusion of such data for the research purposes set out in this protocol is justified as it serves a major public interest, and the processing of data is necessary for the purposes concerned. Climate change, and in particular, rapidly escalating temperatures and heat waves, are a major threat to health across the world, and especially in Africa. Many parts of Africa are warming at twice the average warming rate. Additionally, there is little research on the impacts of extreme heat on pregnant women and children in Africa and this information is vital for developing services to assist these population groups as temperatures escalate. There is thus a major public interest to understand the implications of high temperatures on health in Africa. This information will help to inform the allocation of funding to ameliorate these impacts and assist in selecting appropriate interventions in this regard. Climate change is one of the greatest threats to health in the 21st century, and indeed is an existential threat to the whole globe. Studies such as that described here are essential for addressing this threat.

Data sharing compliance

The data provider will be required to provide a contractual assurance in the data sharing agreement that informed consent procedures were followed for the original studies and that they have individual participant consent to share the data for further scientific research of this nature, in compliance with all data protection legislation applying in the host country.

Considerations specific to data source

As described in Section 4.3, data will be sought from various sources, such as data repositories, from research sites run by Wits University, from other universities or research organisations in South Africa and across sub-Saharan Africa. The ethical and legal considerations for each study may vary by data source, as presented in Table 8 below

Table 2: Ethical and legal considerations for different types of data

Types of data sources			**Approach**	
Type	**Description**	**Example**	**Ethical**	**Legal**
Open source	Available online for download, no data sharing agreements required	Reassessing pregnancy intention and its relation to maternal, perinatal and neonatal outcomes in a low-income setting: A cohort study. https://rdr.ucl.ac.uk/articles/dataset/Pregnancy_intention_and_perinatal_outcomes/8949050	- No further ethics submissions or approvals required - Data will be treated with equal protection standards as other datasets	- Assess study individually for any terms and conditions of sharing (may have embargo's on duration of access etc.) - If publicly available online with data subject consent, data can be transferred subject to country specific legal requirements
In data repository	Access restricted and study staff need to request data, which is reviewed prior to approval	Worldwide Antimalarial Resistance Network	- The data provider will be responsible for their own ethics committee submission, if required	- Complete data requests forms as required by data repository - Complete DSA if required - Adhere to terms and conditions of sharing (e.g., time embargo) - Data transfer is subject to country specific legal requirements
Wits University (including	Participants have signed broad informed consent	ADVANCE study	- No further ethics submissions or approvals required - Participants have agreed to sharing data	- DSA to be completed - POPIA compliance

Wits Health Consortium)			- Ethics committee to be informed about the study contributing data and provided with related documentation	
	Narrow consent given		- Explore feasibility of reconsenting participants - If reconsenting not possible, apply for informed consent waiver to the UCT Faculty of Science Research Ethics Committee	- DSA to be completed - POPIA compliance
South African university/organisation (other than Wits)		SAFE Passage – large cohort from Stellenbosch University	- Assess if participants have signed broad consent - Explore feasibility of reconsenting participants - If reconsenting not possible, apply for informed consent waiver to the UCT Faculty of Science Research Ethics Committee	- DSA to be completed - POPIA compliance
Study and/or PI is from another country	Study was conducted in an African country that is not South Africa – can be single- or multisite study	Kesho Bora study: Maternal ART during pregnancy and breastfeeding prevents more infections than short-course prophylaxis. Study conducted in Kenya, Burkina Faso, and South Africa	- Data provider will be responsible for ethics submission to their own ethics department, if required - Explore informed consent processes	- DSA - POPIA compliance - Compliance with personal data legislation in other countries (data provider is responsible), including cross-border data transfer laws

This DMP (Annex 6) provides additional detail on the steps in which ethical and legal standards will be upheld in the acquisition, handling, and safekeeping of data. As the data will be transferred to a server at UCT, the Faculty of Science Research Ethics Committee will be approached for any additional study specific ethics approvals, if required. As additional databases become available to the study team and DSAs are agreed to, we will notify the UCT Science Faculty Ethics Committee in writing of the new study that has agreed to contribute data to the study. We will provide information about the database prior to actual data transfer. We will inform the committee of the name of the study, study acronym, contact details of the data providers, details of the ethics approval of the original study including informed consent parameters, where available, study country and provide a copy of the DSA. We will make additional information available on the study where requested by the ethics committee. The University of the Witwatersrand Ethics committee will be provided with documentation of the studies that have been included in the IPD and documentation of the decisions made by the UCT Ethics Committee in this regard, six-monthly, or sooner where requested.

Risks to the participants in the primary studies

Risks to a participant in this study are minimal as there are no new procedures involved and steps will be taken to protect their personal information and to avert risks of identification, as described in the data management plan.

7) Training, engagement, and capacity building

Coordinated and consistent implementation of health system operations can be effective and cost-efficient but are labour and resource intensive. For the early warning system to decrease the number of heat-health risks among vulnerable populations, there needs to be a parallel effort to ensure response capabilities are developed that take advantage of the alerts. Working with the ministries of health to build a response system that is useful, intuitive, and capable of being implemented for long-term use is critical. High-risk maps can be translated and shared with partner countries to be integrated into appropriate communication and adaptation planning mechanisms.

Through the Training and Engagement Core (TEC), the HE²AT Center will serves as a Center of Excellence and resource for the DS-I Africa Research Training Program. The HE²AT Center will engage with the communities in which the research is conducted, to increase acceptability of the study and its objectives. These activities use existing engagement mechanisms, engaging with local Community Advisory Boards and media and distributing newsletters. Because the solutions developed by the Center address two major concerns for policymakers (how to warn people about a heatwave and then to track its impacts), we anticipate that local, national and international policymakers will be willing to engage.

The specific aims of the TEC are to:

- Advance knowledge and skills of early career researchers and scientists of the HE²AT Centre and other Research Hubs on transdisciplinary heat-health research using data science approaches
- Build knowledge, skills, and competencies in solutions for addressing heat-health impacts, particularly early warning and response systems
- Develop and implement a systematic strategy for forming, maintaining, and strengthening engagement with governments, industry, non-governmental and related organizations, and communities
- Disseminate results through scientific publications and presentations at scientific meetings, community engagement, policy engagement, and reports to NIH

Approach

The TEC engagement plan will take a phased approach to prioritize different stakeholder groups at different stages of the program lifecycle. Given some of the potential research outputs, it will be important to think about engagement across the implementation phases (e.g., pre- mid – post) – with a focus on the pre-implementation stage in Year 1 and 2, with mid-level implementation starting in Year 3. Across RP2 there are several points of engagement needed, ranging from global to local levels (Figure 4). The TEC will aim to address needs across multiple scales during the lifespan of the development of the heat-health early warning system.

Utilizing the working group for policy and stakeholder engagement the TEC will provide targeted support to the RP2 team to help meet the specific needs of end-users, including linking research outputs to policy interventions and building awareness through the documentation of health impacts of extreme heat. A secondary goal of the engagement working group will be to foster learning opportunities across HE²AT Centre teams to strengthen knowledge engagement in the context of climate change and health. This may include inclusion of presentations to the technical talk series

focusing on engagement approaches, as well as journal clubs within/across working groups to discuss relevant literature.



Figure 4 Levels of stakeholder engagement

The TEC and engagement working group will provide technical support to the activities of research project 2.

- Conducting a stakeholder mapping exercise and workshop to identify and assess needs of decision-makers and end-users involved in the development and implementation of the heat-health early warning system
- Conducting a heat-health policy review and analysis for selected study locations
- Conducting a technical needs assessment and training for health workers to build awareness of heat-health risks, as well as identify, prioritize, and implement intervention strategies based on forecasts and warnings.
- Supporting the engagement with communities and decision-makers to describe the study goals, and gather inputs on the development of the heat-health early warning, including potential intervention strategies.
- Conducting capacity building exercises with key stakeholders including end-users and decision-makers to ensure core competencies are established and knowledge increased to utilize the heat-health early warning system information and implement response actions

8) Dissemination and publication plan

The HE²AT Center will disseminate study findings and other learnings in a timely manner across our consortium, partners, policymakers, the scientific community, other stakeholders, and the general public to maximise its impact. Our engagement plan includes:

- Scientific publications and presentations to scientific conferences
- Community engagement; and
- Policy engagement

Scientific publications from the Research Studies will be published in open access journals with data sharing.

The final protocol for Research Project 2 will be made available on open access to allow replication of the approach in other settings and will be registered with the appropriate registry. It is anticipated that the study will generate high quality data sufficient for at least four publications in high-impact journals. A publication plan will be developed within the consortium that will define the potential publications, authorship, and timeline for manuscript production, as set out in the Administration Core.

Each year there will be *presentations to scientific conferences*, such as the International Society for Environmental Epidemiology, the NIH biannual meetings, the EDCTP Forum, and others. Presentations will also be made upon request to meetings organised by the African Regional Office of the World Health Organization, the United Nations Environment Programme, the United Nations Development Programme, and others. Further, study findings and communication materials may be shared via the Global Heat Health Information Network. The emphasis will be on promoting junior researchers to increase their confidence and skill in preparing and delivering scientific presentations. Costs for these presentations are included in the Research Studies.

The HE²AT Center will promote the Studies and their findings among the *communities* in which the research will be conducted, to ensure increased acceptability of the Studies and their objectives, as well as promote awareness of heat-health risks. Dissemination tools such as newsletters, brochures, and introductory study posters, may be distributed among the community. We will emphasise the guiding principles of Good Participatory Practice, including respect, mutual understanding, integrity, transparency, accountability, and autonomy. Some of the team members already engage frequently with the media about climate or extreme heat concerns, which opens up further avenues for the dissemination of findings.

Study results will be disseminated to the *local, provincial, and national authorities*, to inform their Climate Change and Health Adaptation Planning. Very often these authorities do not receive sufficient technical support for the design and implementation of the types of solution that will be developed in Research Study 1 and Research Study 2 (Overall Hub Aim 4).

The Africa CDC will also play a major role in dissemination activities of the HE²AT Center and are fully committed to participate actively in the activities of the Center, including hosting joint dissemination meetings and/or stakeholder consultations. Timely dissemination and clear communication of results is key for updating guidance and for informing sub-national, national, and international policies. The results will contribute to the development and implementation of heat-health early warning and integrated surveillance systems and associated policies, strategies, and interventions to reduce the health risks of extreme heat in Cote d'Ivoire and South Africa, with the potential of broader application for low- and middle-income countries.

9) Collaborations

Partners in the study are drawn from South Africa, Zimbabwe, Côte d'Ivoire and the United States. The institutions names and locations are presented below in Figure 5.

The RP2 brings together all the institutions of the HE²AT Center in efficient and complementary roles. To get the most out of the collaboration each aim will be run as a collaboration between the different partners.

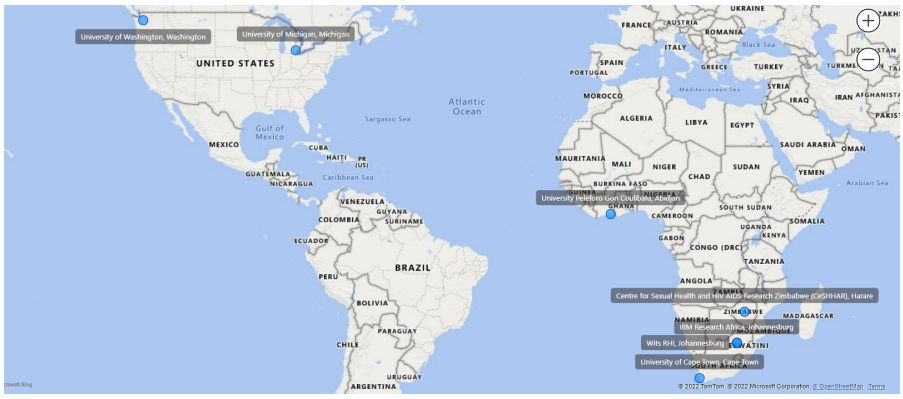


Figure 5 Partners collaborating on the HE²AT study from across Africa and the United States.

10) Main activities, milestones and timelines

Table 3: GANTT chart: milestones and timelines

Milestones / Study quarters	Sept 2021-June 2022				July 2022-June 2023				July 2023-Julne 2024				July 2024-June 2025				July 2025-June 2026				Co*
	1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4	1-4
Research Project 2																					
Initial and continuing IRB and ethics compliance (M6)																					
Training of HE²AT Center on Research Project 2 methods (M6)																					
Protocol paper published (M12)																					
Data analysis plan written (M6)																					
Data harmonization across DSI-Africa platform and other grantees(M12)																					
Biomedical data recoded and merged into one database (M8)																					
Database prepared for vulnerability-heat-health analysis (M12)																					
Publications submitted on primary and secondary study outcomes																					
Engagement with national government for EWS development																					
Stakeholder engagements (community and health departments)																					
Publications submitted on Methodological aspects (M38)																					
Publication on vulnerability-heat-health data																					
Review of District-level surveillance systems performance (M38)																					
PhD enrolment (M7) until award (M54) of one student																					
Dissemination meetings and conferences																					
Final report submitted (M58)																					

* Close-out, M=Month

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Annexes

Annex 1: Databases available to the HE²AT Center

Table 4 Overview of Databases to be used in RP2

Name and source of dataset	Description	UCT /PAIRS	Key variables	Spatio-temporal coverage	Relevance
Biomedical data					
HIV and COVID 19 databases	Pooled health database from multiple large HIV and COVID vaccine trials conducted among adults in Johannesburg, South Africa	WHC studies	Participants are followed up every 3-months for several years,with a multitude of physical measurements, laboratory tests, images and health questionnaires		Research Project 2: The study population has high rates of co- morbidities and adversehealth outcomes
Climate/ weather data					
European Centre for Medium-Range Weather Forecasts (ECMWF) - https://www.ecmwf.int/en/forecasts/datasets/set-i	Outputs from a numerical weather prediction system, run twice daily, designed toproduce state-of-the-art medium (10 days) global forecasts (contains only thelatest forecast)	PAIRS	Temperature (Ground, Min, Max)at 2 m above ground; Solar irradiance; Wind speed (toward east, north) at 10 m above ground; Daily precipitation (total, rate); Dewpoint; Pressure	Spatial: Globalcoverage, 0.065536 deg.Temporal: 3 – 6 hourly & daily res.;Jan 2014 – Oct2019	Determination of heathazard; Thermal comfort metrics; combined climate exposures (forecasts)
IBM TWC (TheWeather Company)Current and historical weather	Data layers from The Weather Company, an IBM Business	PAIRS	Temperature (Change, Min, Max,Feels like); Solar irradiance; Wind (speed, gust & dir.), Rel.Humidity, Daily precipitation (total, rate); Dewpoint; 3-hrlyPressure Change	Spatial: Global coverage, 4km landmass and coastal waterways (hourly & daily res from 2015)	Determination of heathazard; Thermal comfort metrics; combined climate exposures (historical)
Fifth-generation ECMWF high-res. Reanalysis (ERA5) https://cds.climate.copernicus.eu	A global reanalysis dataset combining observed data withthe output of meteorological models. Contains data for 2000, 2005 and from 2009 onwards.	PAIRS	Temperature (2 m above ground,Min, Max); Wind speed (toward east, north); Daily precipitation (total, rate, type); Atmospheric water/ water vapor content, Thermal radiation; Soil temperature; Vegetation types and cover (high, low)	Spatial: Global coverage, 0.131072 degreesPAIRS resolution (raw: 0.25 deg.) Temporal: hourly; coverage from Jan1980 – Jun 2019	Determination of heathazard; Thermal comfort metrics; combined climate exposures (historical)
Copernicus S2S seasonal forecastdata	Model outputs forecasting climate conditions over the three months following the forecast initialization	UCT	Temperature 2m above ground(min, max), Daily precipitation (total)	Temporal: daily	Seasonal (weeks to 3 months) time horizon forecasting of relevantweather conditions (heat hazard) for early warning

CP4-A (NERCJASMIN)	Very high resolution (4km) simulations of historical and future climate over Africa	UCT	Temperature 2m above ground(min, max), daily precipitation (total), multi-level circulation	Temporal: daily	Dynamical downscalingto support sub-urban temp hazard mapping
CORDEX Africa(ESGF)	Ensemble of dynamically downscaled simulations ofAfrican climate to 50km,25km, and 10km resolution	UCT	Temperature 2m above ground(min, max), daily precipitation (total), multi-level circulation fields	Temporal: daily and sub-daily (6hourly)	Dynamical downscalingof climate to support sub-urban temperature hazard mapping
GHCN station data(NOAA GHCN)	Global archive of daily weather station data	UCT	Temperature 2m above ground,daily precipitation (total)	Temporal: daily,station locations	To support statistical downscaling of temperature hazard
Other relevant data domains					
30 m res Elevation(SRTM) (NASAhttps://www2.jpl.nasa.gov/srtm/)	Global elevation data from the Shuttle Radar Topography Mission (SRTM).	PAIRS	Elevation	Released in 2013	Determination of heathazard; Urban heat Island Effect
High res imagery (ESA Sentinel 2) European Space Agency https://sentinel.esa.int/web/sentinel/sentinel-data-access	Images from the Sentinel 2 satellite pair which view land surface regions in 13 spectral bands.	PAIRS	Urban land cover – vegetation coverage, morphological features, possibly pollution levels(AOT). Bands 4 (red), 8 (NIR)and SCL (Scene Classification); Aerosol Optical Thickness; NDVI-sh layer.	Spatial: Global coverage; 0.000064 deg res Temporal: every 5days or faster; from Aug 2015 –Nov 2020.	if there is requirement to control for pollution effects or to look at combined heat-pollutionexposures
Global population (SEDAC) - Gridded Population of the World (GPW), v4 https://sedac.ciesin.columbia.edu/data/collection/gpw-v4	Distribution of human population (counts and densities) on continuous global raster surface. Input data are extrapolated toproduce population estimatesfor 5-year intervals	PAIRS	Population counts and densityestimates	Spatial: Global coverage, 1km grid res Temporal: 5- yearly; coverage from Jan 2000 toJan 2020	Accounting for the population exposed
News coverage (https://www.gdeltproject.org/)	GDELT; Portion of news coverage about specific area and time related to Covid-19.	PAIRS	Global events derived from worldwide news coverage.	Spatial: Globalcoverage,0.008192 deg. res	Example for productionof spatial data layer for news events

Annex 2: Characteristics of the studies

These are the characteristics of each of the trials or cohorts that are included in the study

Revised Cochrane risk-of-bias tool for randomized trials (RoB 2) [82]

Revised Cochrane risk-of-bias tool for randomized trials (RoB 2) TEMPLATE FOR COMPLETION

Edited by Julian PT Higgins, Jelena Savović, Matthew J Page, Jonathan AC Sterne
on behalf of the RoB2 Development Group

Version of 22 August 2019

The development of the RoB 2 tool was supported by the MRC Network of Hubs for Trials Methodology Research (MR/L004933/2- N61), with the support of the host MRC ConDuCT-II Hub (Collaboration and innovation for Difficult and Complex randomised controlled Trials In Invasive procedures - MR/K025643/1), by MRC research grant MR/M025209/1, and by a grant from The Cochrane Collaboration.

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Study details

Reference	

Study design

X Individually-randomized parallel-group trial
☐ Cluster-randomized parallel-group trial
☐ Individually randomized cross-over (or other matched) trial

For the purposes of this assessment, the interventions being compared are defined as

Experimental: _____ Comparator: _____

Specify which outcome is being assessed for risk of bias _____

Specify the numerical result being assessed. In case of multiple alternative analyses being presented, specify the numeric result (e.g. RR = 1.52 (95% CI 0.83 to 2.77) and/or a reference (e.g. to a table, figure or paragraph) that uniquely defines the result being assessed. _____

Is the review team's aim for this result…?

☐ to assess the effect of *assignment to intervention* (the 'intention-to-treat' effect)
☐ to assess the effect of *adhering to intervention* (the 'per-protocol' effect)

If the aim is to assess the effect of *adhering to intervention*, select the deviations from intended intervention that should be addressed (at least one must be checked):

☐ occurrence of non-protocol interventions

☐ failures in implementing the intervention that could have affected the outcome
☐ non-adherence to their assigned intervention by trial participants

Which of the following sources were <u>obtained</u> to help inform the risk-of-bias assessment? (tick as many as apply)
☐ Journal article(s) with results of the trial
☐ Trial protocol
☐ Statistical analysis plan (SAP)
☐ Non-commercial trial registry record (e.g. ClinicalTrials.gov record)
☐ Company-owned trial registry record (e.g. GSK Clinical Study Register record)
☐ "Grey literature" (e.g. unpublished thesis)
☐ Conference abstract(s) about the trial
☐ Regulatory document (e.g. Clinical Study Report, Drug Approval Package)
☐ Research ethics application
☐ Grant database summary (e.g. NIH RePORTER or Research Councils UK Gateway to Research)
☐ Personal communication with trialist
☐ Personal communication with the sponsor

Risk of bias assessment

Responses <u>underlined in green</u> are potential markers for low risk of bias, and responses in red are potential markers for a risk of bias. Where questions relate only to sign posts to other questions, no formatting is used.

Domain 1: Risk of bias arising from the randomization process

Signalling questions	Comments	Response options
1.1 Was the allocation sequence random?		<u>Y / PY</u> / PN / N / NI
1.2 Was the allocation sequence concealed until participants were enrolled and assigned to interventions?		<u>Y / PY</u> / PN / N / NI

1.3 Did baseline differences between intervention groups suggest a problem with the randomization process?		Y / PY / PN / N / NI
Risk-of-bias judgement		Low / High / Some concerns
Optional: What is the predicted direction of bias arising from the randomization process?		NA / Favours experimental / Favours comparator / Towards null /Away from null / Unpredictable

Domain 2: Risk of bias due to deviations from the intended interventions (*effect of assignment to intervention*)

Signalling questions	Comments	Response options
2.1. Were participants aware of their assigned intervention during the trial?		Y / PY / PN / N / NI
2.2. Were carers and people delivering the interventions aware of participants' assigned intervention during the trial?		Y / PY / PN / N / NI
2.3. If Y/PY/NI to 2.1 or 2.2: Were there deviations from the intended intervention that arose because of the trial context?		NA / Y / PY / PN / N / NI
2.4 If Y/PY to 2.3: Were these deviations likely to have affected the outcome?		NA / Y / PY / PN / N / NI
2.5. If Y/PY/NI to 2.4: Were these deviations from intended intervention balanced between groups?		NA / Y / PY / PN / N / NI
2.6 Was an appropriate analysis used to estimate the effect of assignment to intervention?		Y / PY / PN / N / NI

2.7 If N/PN/NI to 2.6: Was there potential for a substantial impact (on the result) of the failure to analyse participants in the group to which they were randomized?		NA / Y / PY / PN / N / NI
Risk-of-bias judgement		Low / High / Some concerns
Optional: What is the predicted direction of bias due to deviations from intended interventions?		NA / Favours experimental / Favours comparator / Towards null /Away from null / Unpredictable

Domain 2: Risk of bias due to deviations from the intended interventions (*effect of adhering to intervention*)

Signalling questions	Comments	Response options
2.1. Were participants aware of their assigned intervention during the trial?		Y / PY / PN / N / NI
2.2. Were carers and people delivering the interventions aware of participants' assigned intervention during the trial?		Y / PY / PN / N / NI
2.3. [If applicable:] If Y/PY/NI to 2.1 or 2.2: Were important non-protocol interventions balanced across intervention groups?		NA / Y / PY / PN / N / NI
2.4. [If applicable:] Were there failures in implementing the intervention that could have affected the outcome?		NA / Y / PY / PN / N / NI
2.5. [If applicable:] Was there non-adherence to the assigned intervention regimen that could have affected participants' outcomes?		NA / Y / PY / PN / N / NI
2.6. If N/PN/NI to 2.3, or Y/PY/NI to 2.4 or 2.5: Was an appropriate analysis used to estimate the effect of adhering to the intervention?		NA / Y / PY / PN / N / NI

Risk-of-bias judgement		Low / High / Some concerns
Optional: What is the predicted direction of bias due to deviations from intended interventions?		NA / Favours experimental / Favours comparator / Towards null /Away from null / Unpredictable

Domain 3: Missing outcome data

Signalling questions	**Comments**	**Response options**
3.1 Were data for this outcome available for all, or nearly all, participants randomized?		Y / PY / PN / N / NI
3.2 If N/PN/NI to 3.1: Is there evidence that the result was not biased by missing outcome data?		NA / Y / PY / PN / N
3.3 If N/PN to 3.2: Could missingness in the outcome depend on its true value?		NA / Y / PY / PN / N / NI
3.4 If Y/PY/NI to 3.3: Is it likely that missingness in the outcome depended on its true value?		NA / Y / PY / PN / N / NI
Risk-of-bias judgement		Low / High / Some concerns
Optional: What is the predicted direction of bias due to missing outcome data?		NA / Favours experimental / Favours comparator /

		Towards null /Away from null / Unpredictable

Domain 4: Risk of bias in measurement of the outcome

Signalling questions	Comments	Response options
4.1 Was the method of measuring the outcome inappropriate?		Y / PY / PN / N / NI
4.2 Could measurement or ascertainment of the outcome have differed between intervention groups?		Y / PY / PN / N / NI
4.3 If N/PN/NI to 4.1 and 4.2: Were outcome assessors aware of the intervention received by study participants?		NA / Y / PY / PN / N / NI
4.4 If Y/PY/NI to 4.3: Could assessment of the outcome have been influenced by knowledge of intervention received?		NA / Y / PY / PN / N / NI
4.5 If Y/PY/NI to 4.4: Is it likely that assessment of the outcome was influenced by knowledge of intervention received?		NA / Y / PY / PN / N / NI
Risk-of-bias judgement		Low / High / Some concerns
Optional: What is the predicted direction of bias in measurement of the outcome?		NA / Favours experimental / Favours comparator / Towards null /Away from null / Unpredictable

Domain 5: Risk of bias in selection of the reported result

Signalling questions	Comments	Response options
5.1 Were the data that produced this result analysed in accordance with a pre-specified analysis plan that was finalized before unblinded outcome data were available for analysis?		Y / PY / PN / N / NI
Is the numerical result being assessed likely to have been selected, on the basis of the results, from...		
5.2. ... multiple eligible outcome measurements (e.g. scales, definitions, time points) within the outcome domain?		Y / PY / PN / N / NI
5.3 ... multiple eligible analyses of the data?		Y / PY / PN / N / NI
Risk-of-bias judgement		Low / High / Some concerns
Optional: What is the predicted direction of bias due to selection of the reported result?		NA / Favours experimental / Favours comparator / Towards null /Away from null / Unpredictable

Overall risk of bias

Risk-of-bias judgement		Low / High / Some concerns
Optional: What is the overall predicted direction of bias for this outcome?		NA / Favours experimental / Favours comparator / Towards null /Away from null / Unpredictable



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The Risk Of Bias In Non-randomized Studies – of Interventions (ROBINS-I) assessment tool

(version for cohort-type studies)

Version 19 September 2016



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ROBINS-I tool (Stage I): At protocol stage

Specify the review question

Participants	
Experimental intervention	
Comparator	
Outcomes	

List the confounding domains relevant to all or most studies

List co-interventions that could be different between intervention groups and that could impact on outcomes

ROBINS-I tool (Stage II): For each study

Specify a target randomized trial specific to the study

Design	Individually randomized / Cluster randomized / Matched (e.g. cross-over)
Participants	
Experimental intervention	
Comparator	

Is your aim for this study…?

☐ to assess the effect of *assignment to* intervention
☐ to assess the effect of *starting and adhering to* intervention

Specify the outcome

Specify which outcome is being assessed for risk of bias (typically from among those earmarked for the Summary of Findings table). Specify whether this is a proposed benefit or harm of intervention.

Specify the numerical result being assessed

In case of multiple alternative analyses being presented, specify the numeric result (e.g. RR = 1.52 (95% CI 0.83 to 2.77) and/or a reference (e.g. to a table, figure or paragraph) that uniquely defines the result being assessed.

Preliminary consideration of confounders

Complete a row for each important confounding domain (i) listed in the review protocol; and (ii) relevant to the setting of this particular study, or which the study authors identified as potentially important.

"Important" confounding domains are those for which, in the context of this study, adjustment is expected to lead to a clinically important change in the estimated effect of the intervention. "Validity" refers to whether the confounding variable or variables fully measure the domain, while "reliability" refers to the precision of the measurement (more measurement error means less reliability).

(i) Confounding domains listed in the review protocol				
Confounding domain	Measured variable(s)	Is there evidence that controlling for this variable was unnecessary?*	Is the confounding domain measured validly and reliably by this variable (or these variables)?	OPTIONAL: Is failure to adjust for this variable (alone) expected to favour the experimental intervention or the comparator?
			Yes / No / No information	Favour experimental / Favour comparator / No information

(ii) Additional confounding domains relevant to the setting of this particular study, or which the study authors identified as important				
Confounding domain	Measured variable(s)	Is there evidence that controlling for this variable was unnecessary?*	Is the confounding domain measured validly and reliably by this variable (or these variables)?	OPTIONAL: Is failure to adjust for this variable (alone) expected to favour the experimental intervention or the comparator?
			Yes / No / No information	Favour experimental / Favour comparator / No information

* In the context of a particular study, variables can be demonstrated not to be confounders and so not included in the analysis: (a) if they are not predictive of the outcome; (b) if they are not predictive of intervention; or (c) because adjustment makes no or minimal difference to the estimated effect of the primary parameter. Note that "no statistically significant association" is not the same as "not predictive".

Preliminary consideration of co-interventions

Complete a row for each important co-intervention (i) listed in the review protocol; and (ii) relevant to the setting of this particular study, or which the study authors identified as important.

"Important" co-interventions are those for which, in the context of this study, adjustment is expected to lead to a clinically important change in the estimated effect of the intervention.

(i) Co-interventions listed in the review protocol		
Co-intervention	Is there evidence that controlling for this co-intervention was unnecessary (e.g. because it was not administered)?	Is presence of this co-intervention likely to favour outcomes in the experimental intervention or the comparator
		Favour experimental / Favour comparator / No information
		Favour experimental / Favour comparator / No information
		Favour experimental / Favour comparator / No information

		Favour experimental / Favour comparator / No information

(ii) Additional co-interventions relevant to the setting of this particular study, or which the study authors identified as important		
Co-intervention	Is there evidence that controlling for this co-intervention was unnecessary (e.g. because it was not administered)?	Is presence of this co-intervention likely to favour outcomes in the experimental intervention or the comparator
		Favour experimental / Favour comparator / No information
		Favour experimental / Favour comparator / No information
		Favour experimental / Favour comparator / No information
		Favour experimental / Favour comparator / No information

Risk of bias assessment

Responses underlined in green are potential markers for low risk of bias, and responses in red are potential markers for a risk of bias. Where questions relate only to sign posts to other questions, no formatting is used.

Signalling questions	**Description**	**Response options**
Bias due to confounding		
1.1 Is there potential for confounding of the effect of intervention in this study? **If N/PN to 1.1:** the study can be considered to be at low risk of bias due to confounding and no further signalling questions need be considered		Y / PY / PN / N
If Y/PY to 1.1: determine whether there is a need to assess time-varying confounding:		

1.2. Was the analysis based on splitting participants' follow up time according to intervention received? **If N/PN**, answer questions relating to baseline confounding (1.4 to 1.6) **If Y/PY**, go to question 1.3.		NA / Y / PY / PN / N / NI
1.3. Were intervention discontinuations or switches likely to be related to factors that are prognostic for the outcome? **If N/PN**, answer questions relating to baseline confounding (1.4 to 1.6) **If Y/PY**, answer questions relating to both baseline and time-varying confounding (1.7 and 1.8)		NA / Y / PY / PN / N / NI

Questions relating to baseline confounding only		
1.4. Did the authors use an appropriate analysis method that controlled for all the important confounding domains?		NA / Y / PY / PN / N / NI
1.5. **If Y/PY to 1.4**: Were confounding domains that were controlled for measured validly and reliably by the variables available in this study?		NA / Y / PY / PN / N / NI
1.6. Did the authors control for any post-intervention variables that could have been affected by the intervention?		NA / Y / PY / PN / N / NI
Questions relating to baseline and time-varying confounding		
1.7. Did the authors use an appropriate analysis method that controlled for all the important confounding domains and for time-varying confounding?		NA / Y / PY / PN / N / NI

1.8. **If Y/PY to 1.7**: Were confounding domains that were controlled for measured validly and reliably by the variables available in this study?		NA / Y / PY / PN / N / NI
Risk of bias judgement		Low / Moderate / Serious / Critical / NI
Optional: What is the predicted direction of bias due to confounding?		Favours experimental / Favours comparator / Unpredictable

Bias in selection of participants into the study		
2.1. Was selection of participants into the study (or into the analysis) based on participant characteristics observed after the start of intervention? **If N/PN to 2.1:** go to 2.4		Y / PY / PN / N / NI
2.2. **If Y/PY to 2.1**: Were the post-intervention variables that influenced selection likely to be associated with intervention?		NA / Y / PY / PN / N / NI
2.3 **If Y/PY to 2.2**: Were the post-intervention variables that influenced selection likely to be influenced by the outcome or a cause of the outcome?		NA / Y / PY / PN / N / NI
2.4. Do start of follow-up and start of intervention coincide for most participants?		Y / PY / PN / N / NI
2.5. **If Y/PY to 2.2 and 2.3, or N/PN to 2.4**: Were adjustment techniques used that are likely to correct for the presence of selection biases?		NA / Y / PY / PN / N / NI
Risk of bias judgement		Low / Moderate / Serious / Critical / NI

Optional: What is the predicted direction of bias due to selection of participants into the study?		Favours experimental / Favours comparator / Towards null /Away from null / Unpredictable

Bias in classification of interventions		
3.1 Were intervention groups clearly defined?		Y / PY / PN / N / NI
3.2 Was the information used to define intervention groups recorded at the start of the intervention?		Y / PY / PN / N / NI
3.3 Could classification of intervention status have been affected by knowledge of the outcome or risk of the outcome?		Y / PY / PN / N / NI
Risk of bias judgement		Low / Moderate / Serious / Critical / NI
Optional: What is the predicted direction of bias due to classification of interventions?		Favours experimental / Favours comparator / Towards null /Away from null / Unpredictable

Bias due to deviations from intended interventions		
If your aim for this study is to assess the effect of assignment to intervention, answer questions 4.1 and 4.2		
4.1. Were there deviations from the intended intervention beyond what would be expected in usual practice?		Y / PY / PN / N / NI
4.2. **If Y/PY to 4.1**: Were these deviations from intended intervention unbalanced between groups *and* likely to have affected the outcome?		NA / Y / PY / PN / N / NI
If your aim for this study is to assess the effect of starting and adhering to intervention, answer questions 4.3 to 4.6		

4.3. Were important co-interventions balanced across intervention groups?		Y / PY / PN / N / NI
4.4. Was the intervention implemented successfully for most participants?		Y / PY / PN / N / NI
4.5. Did study participants adhere to the assigned intervention regimen?		Y / PY / PN / N / NI
4.6. **If N/PN to 4.3, 4.4 or 4.5**: Was an appropriate analysis used to estimate the effect of starting and adhering to the intervention?		NA / Y / PY / PN / N / NI
Risk of bias judgement		Low / Moderate / Serious / Critical / NI
Optional: What is the predicted direction of bias due to deviations from the intended interventions?		Favours experimental / Favours comparator / Towards null /Away from null / Unpredictable

Bias due to missing data		
5.1 Were outcome data available for all, or nearly all, participants?		Y / PY / PN / N / NI
5.2 Were participants excluded due to missing data on intervention status?		Y / PY / PN / N / NI
5.3 Were participants excluded due to missing data on other variables needed for the analysis?		Y / PY / PN / N / NI
5.4 **If PN/N to 5.1, or Y/PY to 5.2 or 5.3**: Are the proportion of participants and reasons for missing data similar across interventions?		NA / Y / PY / PN / N / NI
5.5 **If PN/N to 5.1, or Y/PY to 5.2 or 5.3**: Is there evidence that results were robust to the presence of missing data?		NA / Y / PY / PN / N / NI

Risk of bias judgement		Low / Moderate / Serious / Critical / NI
Optional: What is the predicted direction of bias due to missing data?		Favours experimental / Favours comparator / Towards null /Away from null / Unpredictable

Bias in measurement of outcomes		
6.1 Could the outcome measure have been influenced by knowledge of the intervention received?		Y / PY / PN / N / NI
6.2 Were outcome assessors aware of the intervention received by study participants?		Y / PY / PN / N / NI
6.3 Were the methods of outcome assessment comparable across intervention groups?		Y / PY / PN / N / NI
6.4 Were any systematic errors in measurement of the outcome related to intervention received?		Y / PY / PN / N / NI
Risk of bias judgement		Low / Moderate / Serious / Critical / NI
Optional: What is the predicted direction of bias due to measurement of outcomes?		Favours experimental / Favours comparator / Towards null /Away from null / Unpredictable

Bias in selection of the reported result		
Is the reported effect estimate likely to be selected, on the basis of the results, from... 7.1. ... multiple outcome *measurements* within the outcome domain?		Y / PY / PN / N / NI

7.2 ... multiple *analyses* of the intervention-outcome relationship?		Y / PY / PN / N / NI
7.3 ... different *subgroups*?		Y / PY / PN / N / NI
Risk of bias judgement		Low / Moderate / Serious / Critical / NI
Optional: What is the predicted direction of bias due to selection of the reported result?		Favours experimental / Favours comparator / Towards null /Away from null / Unpredictable

Overall bias		
Risk of bias judgement		Low / Moderate / Serious / Critical / NI
Optional: What is the overall predicted direction of bias for this outcome?		Favours experimental / Favours comparator / Towards null /Away from null / Unpredictable

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Annex 3: Individual-level variables

These are the variables that were collected on individual participants at the enrolment or follow-up visits.

Table 5 Variable List for RP2

Variable Name	Variable Abbreviation	Description
Date	date	The particular day, month and year an event has happened or will happen.
Time	time (local)	The continuum of experience in which events pass from the future through the present to the past.
Country	country	A collective generic term that refers here to a wide variety of dependencies, areas of special sovereignty, uninhabited islands, and other entities in addition to the traditional countries or independent states.
Patient Identifier	patient_id	An alphanumeric identifier assigned to a specific patient.
Age-Years	age_enrolment	The length of a person's life, stated in years since birth.
Race	race	A geographic ancestral origin category that is assigned to a population group based mainly on physical characteristics that are thought to be distinct and inherent.
Sex	sex	The assemblage of physical properties or qualities by which male is distinguished from female; the physical difference between male and female; the distinguishing peculiarity of male or female.
Height	height	The vertical measurement or distance from the base to the top of an object; the vertical dimension of extension.
Body Weight	weight	The weight of a subject.
Body Mass Index	bmi	An individual's weight in kilograms divided by the square of the height in meters.
Location	study_location	A position, site, or point in space where something can be found.
Address	participant_address	A standardized representation of the location of a person, business, building, or organization.
Housing Type	housing_type	The classification of a residential structure

Number in household (observable entity)	num_people_household	Number in household
exposure to air conditioning unit	aircon_access	A exposure event involving the interaction of an exposure receptor to the condition of air conditioning unit.
Income	income	A gain or recurrent benefit during a period of time, usually measured in money that derives from capital or labor.
Household Income	house_income	A demographic parameter indicating the amount of earnings made by a family.
Substance Abuse	substance_abuse	Maladaptive pattern of drug or alcohol use that may lead to social, occupational, psychological, or physical problems.
Smoking Status	smoking_status	An indication of a person's current tobacco and nicotine consumption as well as some indication of smoking history.
Alcohol Abuse	alcohol_status	The use of alcoholic beverages to excess, either on individual occasions ("binge drinking") or as a regular practice.
Employment Status	employment	The state of a person with regard to earning wages or salary.
Education Level	education_years	An indication of the years of schooling completed in graded public, private, or parochial schools, and in colleges, universities, or professional schools.
Lost To Follow-Up	loss_to_follow_up	The subject was not available for follow-up procedures.
Hypertension	hypertension	Blood pressure that is abnormally high.
Diabetes Mellitus	dm	A metabolic disorder characterized by abnormally high blood sugar levels due to diminished production of insulin or insulin resistance/desensitization.
Number of Fetuses	number_of_foetuses	A measurement of the total number of fetuses, which includes alive and dead fetuses, present in the uterus.
Death Indicator	death	Death is the permanent cessation of all biological functions that sustain a living, physical organism.

Hospitalization	hospital_admission	The confinement of a patient in a hospital.
Gastroenteritis	gastroenteritis	An inflammatory disorder that affects the upper and lower gastrointestinal tract. Most commonly, this is attributed to viruses; however bacteria, parasites or adverse reactions can also be the culprit. Symptoms include acute diarrhea and vomiting.
Pneumonia	pneumonia	An acute, acute and chronic, or chronic inflammation focally or diffusely affecting the lung parenchyma, due to infections (viruses, fungi, mycoplasma, or bacteria), treatment (e.g. radiation), or exposure (inhalation) to chemicals. Symptoms include cough, shortness of breath, fevers, chills, chest pain, headache, sweating, and weakness....
Urinary Tract Infection	urinary_tract_infection	A bacterial infectious process affecting any part of the urinary tract, most commonly the bladder and the urethra. Symptoms include urinary urgency and frequency, burning sensation during urination, lower abdominal discomfort, and cloudy urine.
Group B Streptococcus Infection	groupb_strep_infection	Group B Streptococcus, also known as Streptococcus agalactiae, colonizes the vaginal and gastrointestinal tracts of up to 45% of healthy women and may infect neonates in utero or during delivery, causing neonatal sepsis in 1-2% of colonized neonates. GBS infection may also occur in nonpregnant (particularly elderly) adults with underlying medical conditions, presenting as urinary tract infection, pneumonia, or soft-tissue infection. [NCI]
Syphilis	syphilis_status	A contagious bacterial infection caused by the spirochete Treponema pallidum. It is a sexually transmitted disorder, although it can also be transmitted from the mother to the fetus in utero. Typically, it is initially manifested with a single sore which heals without treatment. If the

		infection is left untreated, the initial stage is followed by skin rash and mucous membrane lesions. A late stage follows, which is characterized by damage of the internal organs, including the nervous system.
HIV Status	hiv_status	The result of testing to determine if an individual is infected with the human immunodeficiency virus.
Hepatitis B Infection	HepB_status	A viral infection caused by the hepatitis B virus.
Schistosomiasis	schistosomiasis_status	A parasitic infection caused by flukes of the genus Schistosoma. Signs and symptoms include fever, abdominal pain, eosinophilia and hepatosplenomegaly. If left untreated it may eventually cause liver damage leading to cirrhosis, bladder cancer and kidney failure.
Tuberculosis	tb_status	A chronic, recurrent infection caused by the bacterium Mycobacterium tuberculosis. Tuberculosis (TB) may affect almost any tissue or organ of the body with the lungs being the most common site of infection. The clinical stages of TB are primary or initial infection, latent or dormant infection, and recrudescent or adult-type TB. Ninety to 95% of primary TB infections may go unrecognized. Histopathologically, tissue lesions consist of granulomas which usually undergo central caseation necrosis. Local symptoms of TB vary according to the part affected; acute symptoms include hectic fever, sweats, and emaciation; serious complications include granulomatous erosion of pulmonary bronchi associated with hemoptysis. If untreated, progressive TB may be associated with a high degree of mortality. This infection is frequently observed in immunocompromised individuals with AIDS or a history of illicit IV drug use.
Hemoglobin	haemoglobin	The red respiratory protein of erythrocytes, consisting of

		approximately 3.8% heme and 96.2% globin (64.5 KD), which as oxyhemoglobin (HbO2) transports oxygen from the lungs to the tissues where the oxygen is readily released and HbO2 becomes Hb.
Creatinine	creat	The breakdown product of creatine, a constituent of muscle tissue, that is excreted by the kidney and whose serum level is used to evaluate kidney function.
Creatinine Clearance	creat_clearance	The determination of the clearance of endogenous creatinine, used for evaluating the glomerular filtration rate.
HIV Viral Load Measurement	hiv_vl	The determination of the HIV viral load in a specimen.
CD4 Expressing Cell Count	cd4	The determination of the amount of the CD4 expressing cells in a sample.
Platelet Count	plt	The determination of the number of platelets in a biospecimen.
Aspartate Aminotransferase Measurement	ast	A quantitative measurement of aspartate aminotransferase present in a sample.
Alanine Aminotransferase Measurement	alt	A quantitative measurement of alanine aminotransferase present in a sample.
Protein to Creatinine Ratio Measurement	urine_pcr	The determination of the ratio of total protein compared to creatinine present in a sample. The measurement may be expressed as a ratio or percentage.
Alkaline Phosphatase Measurement	alp	A quantitative measurement of alkaline phosphatase present in a sample.
mean corpuscular volume	mcv	A mean corpuscular volume is the result of calculation of the mean volume of erythrocytes in a blood sample.
mean corpuscular hemoglobin concentration	mchc	The mean corpuscular hemoglobin concentration is a measure of the concentration of hemoglobin in a given volume of packed red blood cell

Viral Resistance Domain	hiv_resistance_mutations	A findings domain that captures information regarding the genetics of viral drug resistance. It contains the reference sequence used to validate the observed genetic mutation of interest.
Drug resistance to antiretroviral therapy (disorder)	hiv_drug_resistance	A binary variable describing the presence or abscence of any drug resistance to antiretroviral therapy.
Highly Active Antiretroviral Therapy	art_regimen	Drug therapy which targets retrovirus function by multiple mechanisms.
Noncompliance with antiretroviral medication regimen (finding)	art_adherance	Noncompliance with antiretroviral therapy
Study Subject Radiography Report	xray_findings	Records pertaining to the findings from a study subject's radiographic images.
Oligohydramnios	oligohydramnios	A lower than normal quantity of amniotic fluid in the amniotic sac as compared to normal values. Typically associated with an amniotic fluid index (AFI) of less than 5 cm or a single maximum vertical pocket (MVP) of less than 2 cm.
thyroid stimulating hormone measurement	tsh	Is a quantification of thyroid-stimulating hormone, a glycoprotein and hormone secreted from the pituitary which regulates the thryoid.
Lower Respiratory Tract Infection	lrti	An acute or chronic, viral or bacterial infectious process that affects the lower respiratory tract.
Upper Respiratory Tract Infection	urti	An infectious process affecting the upper respiratory tract (nose, paranasal sinuses, pharynx, larynx, or trachea). Symptoms include congestion, sneezing, coughing, fever, and sore throat.

Direct Bilirubin Measurement	direct_bili	The bilirubin is bound to glucuronide to form conjugated bilirubin (direct bilirubin). Direct Bilirubin measurement is accomplished by a colorimetric method. Direct Bilirubin in biological fluids reacts with sulfanilic acid at acidic pH to produce a red colored complex. The optical density of produced color has a direct relationship with Direct Bilirubin concentration in the solution.
Indirect Bilirubin Measurement	indirect_bili	Bilirubin is in the insoluble form, unconjugated bilirubin (indirect bilirubin).The non-water soluble, free bilirubin does not react with sulfanilic acid at acidic pH to produce a red colored complex until an accelearator, alcohol, is added to the solution to perform a quantitative measurement of unconjugated bilirubin levels.
Amylase Measurement	amylase	A quantitative measurement of amylase present in a sample.
Lipase Measurement	lipase	The determination of the amount of lipase present in a sample.
Cholesterol Measurement	cholesterol	The determination of the amount of total cholesterol present in a sample.
mean corpuscular hemoglobin	mch	The MCH is the average mass of hemoglobin per red blood cell in a sample of blood and is calculated by dividing the total mass of hemoglobin by the RBC count
Low Density Lipoprotein Cholesterol Measurement	ldlc	The determination of the amount of low-density lipoprotein cholesterol present in a sample.
High Density Lipoprotein Cholesterol Measurement	hdlc	The determination of the amount of high-density lipoprotein cholesterol present in a sample.
Glycosylated Hemoglobin Measurement	hba1c	A quantitative measurement of the amount of glycosylated hemoglobin present in a sample of blood.
Albumin Measurement	albumin	A quantitative measurement of albumin present in a sample.
Cortisol Measurement	cortisol	The determination of the amount of cortisol present in a sample.

Whole Parathyroid Hormone Measurement	pth	The determination of the amount of the whole parathyroid hormone (consisting of amino acids 1-84) in a sample.
Free Thyroxine Measurement	ft4	The determination of the amount of free thyroxine present in a sample.
Free Triiodothyronine Measurement	ft3	The determination of the amount of free triiodothyronine present in a sample.
Blood Urea Nitrogen Measurement	urea	A quantitative measurement of the amount of urea nitrogen present in a serum sample.
Calcium Measurement	calcium	A quantitative measurement of the amount of calcium present in a sample.
Bone Mineral Density Z-Score	dexa_scan	A statistical score representing the number of standard deviations above or below what is expected for an individual's bone density based on his age, sex, weight, and race. Z-scores are most useful in evaluating low bone density in children, premenopausal women, and men younger than age fifty.
Serum Uric Acid Measurement	uric_acid	A quantitative measurement of the amount of uric acid present in a sample of serum.
Potassium Measurement	potassium	A quantitative measurement of the amount of potassium present in a sample.
Sodium Measurement	sodium	A quantitative measurement of the amount of sodium present in a sample.
COVID-19 RT-PCR assay	covid_19_pcr	Result of a COVID-19 PCR based assay.
C-Reactive Protein Measurement	crp	A quantitative measurement of the amount of C-reactive protein present in a sample.
Procalcitonin Measurement	pct	The determination of the amount of the procalcitonin in a sample.
interleukin-6 measurement	il-6	Is a quantification of interleukin-6, a pro-inflammatory and anti-inflammatory cytokine.

International Normalized Ratio	inr	A system, commonly called the INR, established by the World Health Organization (WHO) and the International Committee on Thrombosis and Hemostasis for reporting the results of blood coagulation (clotting) tests. All results are standardized using the international sensitivity index (ISI) for the particular thromboplastin reagent and instrument combination utilized to perform the test; the ratio of a patient's clotting time to the lab's mean reference value is normalized against the ISI. (from medterms.com and medicine.ucsf.edu)
D-Dimer Measurement	d_dimers	The determination of the amount of d-dimers present in a sample.
Unipolar Depression	depression	A mood disorder having a clinical course involving one or more episodes of serious psychological depression that last two or more weeks each, do not have intervening episodes of mania or hypomania, and are characterized by a loss of interest or pleasure in almost all activities and by some or all of disturbances of appetite, sleep, or psychomotor functioning, a decrease in energy, difficulties in thinking or making decisions, loss of self-esteem or feelings of guilt, and suicidal thoughts or attempts.
Generalized Anxiety Disorder	gen_anxiety_disorder	An anxiety disorder characterized by excessive and difficult-to-control worry about a number of life situations. The worry is accompanied by restlessness, fatigue, inability to concentrate, irritability, muscle tension, and/or sleep disturbance and lasts for at least 6 months.
Adverse Event	adverse_event	Any unfavorable or unintended disease, sign, or symptom (including an abnormal laboratory finding) that is temporally associated with the use of a medical treatment or procedure, and that may or may not be considered related to the medical treatment or

		procedure. Such events can be related to the intervention, dose, route of administration, patient, or caused by an interaction with another drug(s) or procedure(s)
Heart Rate	heart_rate	The number of heartbeats per unit of time, usually expressed as beats per minute.
Systolic Blood Pressure	systolic_blood_pressure	The maximum pressure exerted into the systemic arterial circulation during the contraction of the left ventricle of the heart.
Diastolic Blood Pressure	diastolic_blood_pressure	The minimum pressure exerted into the systemic arterial circulation during cardiac ventricular relaxation and filling.
Mean Arterial Pressure	mean_arterial_pressure	The mean pressure of the blood within the arterial circulation. The arterial pressure may be directly measured by insertion of an intra-arterial catheter connected to a transducer. The mean arterial pressure (MAP) can be calculated by subsequent analysis of the waveform. MAP can be approximated without an invasive procedure using the following formula: diastolic pressure plus 1/3 of the pulse pressure, where pulse pressure is systolic pressure - diastolic pressure.
Oxygen Saturation Measurement	oxygen_saturation	The measurement of the ratio of oxygenated hemoglobin to total hemoglobin in the blood
Myocardial Infarction	myocardial_infarct	Gross necrosis of the myocardium, as a result of interruption of the blood supply to the area, as in coronary thrombosis.

Annex 4: Land Surface Temperature retrieval workflow

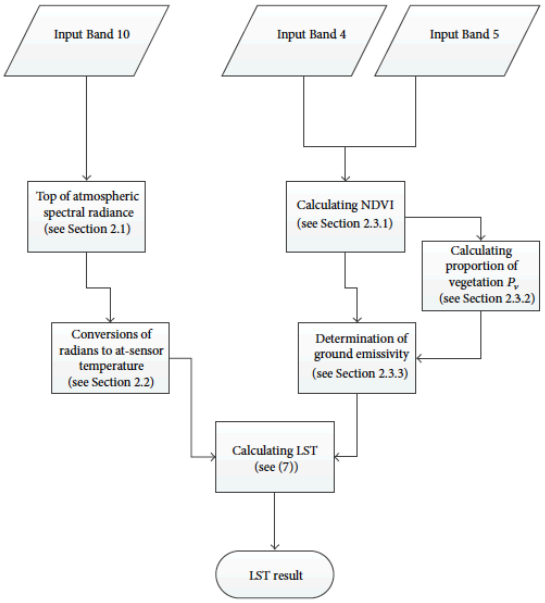


Figure 6 Land Surface Temperature retrieval workflow

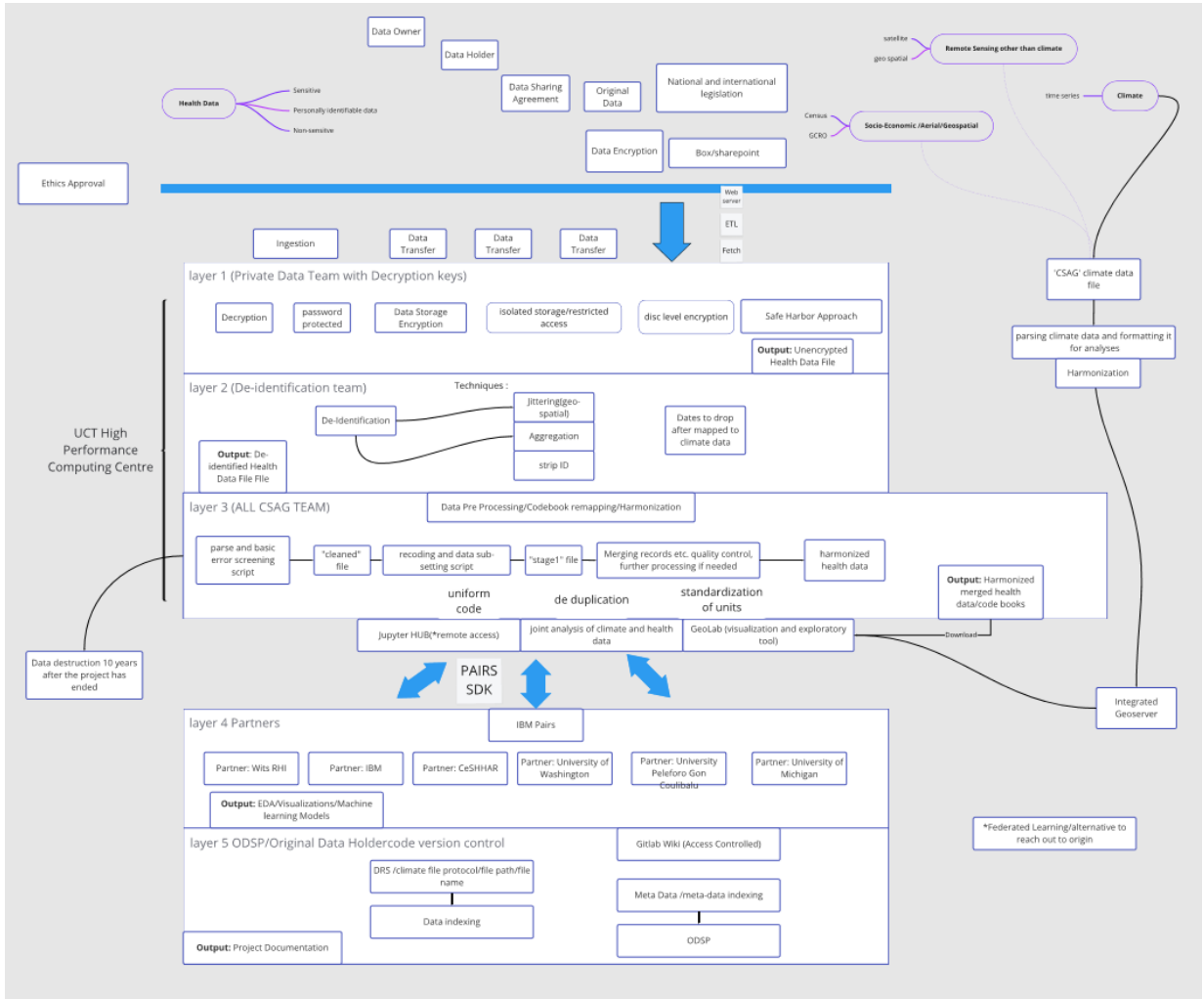
Annex 5: Example Impact Assessment Questions and Indicators

	Evaluation questions	Potential indicators
Alert protocol	Were the alerts issued efficiently?	Frequency of partner notification and public alerts Timeliness of alert information received Timeliness and efficiency of message delivery to the end-users
	Are extreme heat events forecasted and monitored accurately?	Quality of surveillance data Frequency of warnings and alerts issued in relation to actual weather conditions occurring Capacity of participating agencies to monitor and deliver surveillance and weather data
Response plan	How involved were stakeholders with implementing response measures?	Level of participation of agencies and other community groups in education activities, issuing warnings and responding to an alert Number and types of response measures delivered by stakeholders
	Did stakeholders follow the response plan and find it helpful?	Number and diversity of engaged stakeholders and meeting frequency Perceived importance of the heat response among stakeholders Partners' views on the degree of coordination of activities Stakeholders' views on the adequacy of support offered Level of stakeholder satisfaction
	Are response measures being used by the public (e.g. cooling centres)?	Number of at-risk people who took preventive actions Number of people, their demographic makeup, and length of visits to cooling facilities Number of people and their demographic makeup who took advantage of other response measures
Communicatio n plan	Were key messages and services provided to the public?	Number of planned communication elements delivered Vulnerable and general populations reached by each communication element Number and types of inquiries received Number and types of resources distributed Promotion and publicity received through media activities
	Was the target population aware of heat-health EWS and its key messages?	Number of media and information sources engaged as part of the outreach campaign Penetration of key messages into the media Accessibility of information to the public
	Did the target population understand and follow key messages?	Number of at-risk people who perceive extreme heat to be a health risk Number of at-risk people who can identify preventive measures Capacity of targeted population to recall accurate messaging

Operational costs	What resources are used to operate the heat-health EWS?	Resources used by each partner Resources required for collecting and monitoring surveillance data Staff time spent on the program at various stages Costs to communicate messages to stakeholders and the public Costs of maintaining the system

Source: Adapted from [84]

Annex 7: Overall Data Management Flow



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